Sharp normal stability and finite actual travel for reciprocal-clock IGATH-2

Abstract

We study the literal fixed-step inertial gradient recursion with unregularized gradient history and reciprocal Tikhonov regularization near a compact C^5 minimum manifold. A collision-safe rational metric gives the sharp normal band $0 < H < 8$ and a self-contained capture proof with independent-history entry transients. On this common foundation, a moving stable and unstable phase bundle yields finite actual travel under Morse or Morse–Bott geometry, including approached saddle components and all tangent velocity modes. The same actual-map forcing identifies full vector Morse bias, moving-base transverse bias, and exact center and travel-tail constants; a uniform prepared circle realizes the $k^{-1/2}$ tail. A parallel one-sided Kurdyka–Łojasiewicz argument allows degenerate critical sets, using a proved transfer from the restricted squared norm to a cubic-clock energy and giving desingularizer and power tails. At $H = 8$, an exact quadratic example has a positive product-normalized consecutive-pair limit and explicit growth envelopes under both clocks. All capture and selection statements retain the original integer index and their local entry hypotheses.

Keywords: IGATH-2; reciprocal Tikhonov clock; sharp normal stability; finite path length; Morse–Bott geometry; Kurdyka–Łojasiewicz inequality

2020 Mathematics Subject Classification. Primary 37N40; Secondary 39A30, 37C60, 65K10.

1 The literal recursion and capture theorem

Let M be a compact embedded C^5 submanifold without boundary of $\mathbb{R}^d$, and let F be C^5 on an open neighborhood of M. Assume, for every $p \in M$,

$$\nabla F(p) = 0, \qquad \ker \nabla^2 F(p) = T_p M, \qquad \operatorname{spec}(H_p) \subset [\mu, L], \quad 0 < \mu \le L < 8, \tag{1}$$

where $H_p = \nabla^2 F(p)|_{N_p M}$ is the self-adjoint normal restriction. Thus M is a normally nondegenerate manifold of local minima. Different connected components need not have the same critical value. Write P_p, N_p for the orthogonal tangent and normal projectors, respectively. In a sufficiently small tubular neighborhood U, let π denote the nearest-point projection. Norms are Euclidean unless another norm is specified. Set

$$q(p) = \tfrac{1}{2}|p|^2, \qquad g(p) = \operatorname{grad}_M q(p) = P_p p.$$

Our second fundamental form convention is $\mathrm{II}_p(a, b) = N_p D_a b$. Equivalently, the normal component of the acceleration of a geodesic with initial velocity a is $\mathrm{II}_p(a, a)$.

For integers $k \geq K$, start from the independent history x_{K-1}, x_K and use the literal recursion

$$\begin{aligned} y_k &= x_k + \left(1 - \frac{3}{2\sqrt{k}}\right)(x_k - x_{k-1}) - \frac{1}{4}\big(\nabla F(x_k) - \nabla F(x_{k-1})\big), \\ x_{k+1} &= y_k - \frac{1}{4}\left(\nabla F(y_k) + \frac{y_k}{k}\right). \end{aligned} \tag{2}$$

The clock is k, not $k - K$ and not a restarted index. No condition on an earlier prefix is asserted or needed. For $x_k \in U$, define

$$\begin{aligned} p_k &= \pi(x_k), & n_k &= x_k - p_k, \\ u_k &= N_{p_k}(x_k - x_{k-1}), & v_k &= \sqrt{k}\, P_{p_k}(x_k - x_{k-1}). \end{aligned} \tag{3}$$

Exactly,

$$d_k := x_k - x_{k-1} = h_k v_k + u_k, \qquad h_k = k^{-1/2}. \tag{4}$$

The normal history u_k is projected at the *current* base p_k; it is not $n_k - n_{k-1}$.

The map is the IGATH-2 scheme in the final published version of Attouch, Balhag, Chbani, and Riahi [1], Section 4.2, with

$$s = \tfrac{1}{4}, \qquad \varepsilon_k = \tfrac{1}{k}, \qquad \delta = 3, \qquad \beta = \tfrac{1}{2}.$$

Indeed $\delta\sqrt{s\varepsilon_k} = 3/(2\sqrt{k})$ and $\beta\sqrt{s} = 1/4$. The gradient difference is unregularized, and the Tikhonov term acts at y_k. That source introduces the explicit scheme but leaves its convergence analysis for further study. Its continuous and proximal results are not used as a local nonconvex convergence theorem here.

Theorem 1 (Sharp-range original-clock capture)**.** *Under* (1)*, there exist a tubular radius $r > 0$ and a fixed $\theta > 0$, depending only on the displayed geometry and F, with the following property. For every prescribed finite $B_0 \geq 0$, there is an integer $K_0(B_0)$ such that, at every original index $K \geq K_0(B_0)$, every independent history pair satisfying*

$$x_K \in U_r, \qquad \sqrt{K}\,|n_K| \leq \theta, \qquad \sqrt{K}\,|u_K| \leq \theta, \qquad |v_K| \leq B_0 \tag{5}$$

has a well-defined suffix under (2) *for all $k \geq K$. The predecessor x_{K-1} automatically lies in the neighborhood where F is defined once K_0 is chosen; this is a consequence of* (5)*.*

The following conclusions hold.

1. *All x_k and y_k remain in a fixed controlled tubular neighborhood of M, all p_k are defined, and $|v_k| \leq B(B_0)$ for some finite $B(B_0)$.*

2. *For constants $C(B_0) < \infty$ and $0 < \rho < 1$,*

$$|n_k| + |u_k| \leq C(B_0)\theta K^{-1/2}\rho^{k-K} + \frac{C(B_0)}{k}. \tag{6}$$

3. *The slow-clock dissipation satisfies*

$$\sum_{k=K}^{\infty} k^{-1/2}\big(|v_k|^2 + |g(p_k)|^2\big) < \infty, \qquad v_k \longrightarrow 0, \quad g(p_k) \longrightarrow 0. \tag{7}$$

4. *With* $\operatorname{Crit}_M q = \{p \in M : g(p) = 0\}$,

$$\operatorname{dist}(x_k, \operatorname{Crit}_M q) \longrightarrow 0.$$

The nonempty compact omega-limit set of (x_k) *is contained in* $\operatorname{Crit}_M q$. *The predictor sequence* (y_k) *has exactly the same omega-limit set.*

5. *The actual normal position and normal history satisfy*

$$\begin{aligned} n_k &= k^{-1}c(p_k, v_k) + O(k^{-3/2}), \\ u_k &= -\frac{1}{2k}\mathrm{II}_{p_k}(v_k, v_k) + O(k^{-3/2}), \end{aligned} \tag{8}$$

where

$$c(p, v) = -H_p^{-1}N_p p - (4H_p^{-1} - I)\mathrm{II}_p(v, v). \tag{9}$$

In particular,

$$kn_k + H_{p_k}^{-1}N_{p_k}p_k \longrightarrow 0.$$

These are derived tracking estimates, not an assumed exact invariant branch.

The big-O constants may depend on the admitted history and its original entry index. The tube, θ, *and the choice of* K_0 *do not depend on the admitted history pair beyond* B_0 *and the entry bounds.*

The upper bound $L < 8$ is used throughout this manuscript, with constants constructed on that full spectral band. The earlier sufficient range $L < 4$ is a specialization; its constants are not reused on the larger interval. The capture conclusion alone does not imply one-point convergence, finite travel, or global least-norm selection. Morse–Bott geometry gives finite travel and precise bias in Sections 3–4. A one-sided scalar gradient inequality gives a parallel route through degenerate critical sets in Section 5.

The exact normal stability interval is $0 < H < 8$ for this fixed step. Theorem 9 proves failure of a universal bounded-capture assertion at $H = 8$ using the actual nonautonomous recursion, not only its frozen eigenvalues. No assertion about every individual coordinate magnitude is inferred from consecutive-pair growth.

2 Normal stability and actual capture

2.1 Uniform geometry

Put $h_k = k^{-1/2}$, $\gamma = 3/2$, and $b = 1/4$. On N_pM, set

$$A_p = I - bH_p.$$

The operator A_p may have zero or negative eigenvalues. Its eigenvalues belong to the compact interval $[1 - L/4, 1 - \mu/4] \subset (-1, 1)$.

Choose a closed tubular neighborhood of M contained in the domain of F, and a slightly larger such neighborhood still contained in that domain. Shrink their radii as needed below. Compactness and the stated regularity give uniform bounds for the required derivatives of the nearest-point projection, tangent and normal projectors, F, and the intrinsic exponential map on a uniform small geodesic neighborhood of the diagonal. All Taylor remainders below are uniform in $p \in M$. A compact C^5 embedded manifold suffices for these finite-order estimates; neither analyticity nor a Łojasiewicz inequality is used.

We use tangent and normal parallel transport along the unique short minimizing geodesic between sufficiently close points of M. Transport is isometric in each bundle. Bars over vectors based at the next point indicate transport back to the current point. Norms on pairs are Euclidean product norms. Constants $C, C_{R,D}$ and so on are finite and may increase between displays. All constants are chosen before the entry index K.

For $p \in M$ and small $\eta \in N_pM$, Taylor expansion gives

$$\nabla F(p + \eta) = H_p\eta + O(|\eta|^2). \tag{10}$$

Consequently, if $z = r + \eta$ with $r = \pi(z)$, then

$$|P_p\nabla F(z)| \le C\big(|r - p|\,|\eta| + |\eta|^2\big). \tag{11}$$

Indeed $P_rH_r\eta = 0$ and $|P_p - P_r| \le C|p - r|$. For small $\xi \in T_pM$, the geodesic expansion is

$$\exp_p \xi = p + \xi + \tfrac{1}{2}\mathrm{II}_p(\xi, \xi) + O(|\xi|^3). \tag{12}$$

For tangent transport τ from p to r and an ambient vector $a + c$, where $a \in T_pM$ and $c \in N_pM$, one has

$$\tau^{-1}P_r(a + c) = a + O\big(\mathrm{dist}_M(p, r)|c| + \mathrm{dist}_M(p, r)^2|a|\big). \tag{13}$$

To see the absence of a first-order error on a, the transported vector and its orthogonal projection onto T_rM have the same first-order ambient normal term $\mathrm{II}_p(\log_p r, a)$. Their difference is second order. The term from c is bounded by projector variation. Taylor expansion in a finite collection of local orthonormal bundle frames makes the remainder uniform.

Finally, differentiating $\nabla F(\exp_p(tv)) = 0$ twice at $t = 0$ gives

$$D^3F(p)[v, v] = -H_p\mathrm{II}_p(v, v), \qquad v \in T_pM. \tag{14}$$

Here $D^3F(p)[v, v]$ is the vector obtained by contracting two slots of the third derivative. This is a vector identity, not merely its normal projection.

2.2 A finite-history one-step estimate

We work at one original integer k. Let $h = k^{-1/2}$, $p = \pi(x)$, $n = x - p$, and decompose the actual history $d = x - x_{\mathrm{prev}}$ as

$$d = hv + u, \qquad v \in T_pM, \quad u \in N_pM.$$

Fix finite bounds R, D, and assume $|v| \le R$ and $|n| + |u| \le Dh$. If h is sufficiently small depending on R, D, the predecessor, predictor, and next point all belong to the larger chosen tube. Their projections are defined, and $p_+ = \pi(x_+)$ is in the uniform geodesic neighborhood of p. Write

$$z = (n, u), \qquad z_+ = (n_+, u_+), \qquad v_+ = h_+^{-1} P_{p_+}(x_+ - x), \quad h_+ = (k+1)^{-1/2}.$$

The following estimates are uniform:

$$\bar{z}_+ = B_p z + R^N, \qquad |R^N| \le C_{R,D} h^2, \qquad B_p = \begin{pmatrix} A_p & A_p^2 \\ A_p - I & A_p^2 \end{pmatrix}, \tag{15}$$

$$\log_p p_+ = hv + r^p, \qquad |r^p| \le C_{R,D} h^2, \tag{16}$$

$$\bar{v}_+ = v - h\big(\gamma v + bg(p)\big) + r^v, \qquad |r^v| \le C_{R,D} \left(h^2 + |z| + \frac{|z|^2}{h} \right). \tag{17}$$

The bars in (15) and (17) mean normal and tangent transport, respectively.

Proof of the one-step estimates. The point $x_{\mathrm{prev}} = p + n - u - hv$ is $O(h)$ from p. The projection Taylor formula at p bounds its normal displacement from M by $C(|z| + h^2)$, and its projected base is $O(h)$ from p. The analogous assertions for y, x_+ follow initially from their $O(h)$ proximity to p; the refined normal bounds follow from their defining formulas and (10). More explicitly, first-order gradient expansion at p gives

$$\nabla F(x) - \nabla F(x_{\mathrm{prev}}) = H_p u + O(h^2), \qquad y - p = hv + n + A_p u + O(h^2).$$

The normal displacement of y from M is $n + A_p u + O(h^2)$, so

$$x_+ - p = hv + A_p n + A_p^2 u + O(h^2).$$

Subtracting the $O(h^2)$ normal component of $p_+ - p$ gives

$$\bar{n}_+ = A_p n + A_p^2 u + O(h^2).$$

Moreover,

$$u_+ = N_{p_+}\big[(p_+ - p) + (n_+ - n)\big].$$

The first term is $O(h^2)$, and transport and projector changes on n are $O(h|n|)$. Hence

$$\bar{u}_+ = (A_p - I)n + A_p^2 u + O(h^2),$$

which proves (15).

The tangential estimate requires more than an undifferentiated $O(h^2)$ gradient error. By (11), the normal displacement bounds just obtained, and the $O(h)$ base changes,

$$|P_p \nabla F(x)| \le C|n|^2,$$
$$|P_p \nabla F(x_{\text{prev}})| + |P_p \nabla F(y)| \le C_{R,D}\big(h|z| + |z|^2 + h^3\big).$$

Also $P_p y = g(p) + hv + O(h^2)$; here $g(p)$ is the tangent projection of the ambient position p itself. Applying P_p to the actual increment therefore yields

$$\begin{aligned} P_p(x_+ - x) &= hv - \gamma h^2 v - bh^2 g(p) + E, \\ |E| &\le C_{R,D}\big(h^3 + h|z| + |z|^2\big). \end{aligned} \tag{18}$$

This uses the unregularized gradient difference and the Tikhonov term at y exactly as in (2).

The normal component $N_p(x_+ - x)$ is $O(|z| + h^2)$. By (13), replacing the current-base tangent projection of the actual increment by its back-transported next-base tangent projection adds at most $C(h|z| + h^3)$. Division by h_+ and

$$h/h_+ = \sqrt{1 + h^2} = 1 + O(h^2)$$

prove (17).

For position, project $x_+ - x = (p_+ - p) + (n_+ - n)$ tangentially at p. The projection of n is zero; that of n_+ has size at most $Ch(|z| + h^2)$. Formula (12) says that $P_p(p_+ - p)$ differs from $\log_p p_+$ by $O(h^3)$. Equation (18) and $|z| \le Dh$ now prove (16). No continuous-time limit has replaced the actual recursion. ∎

2.3 The exact normal interval and a regular Lyapunov metric

Lemma 1 (Exact scalar Schur interval)**.** *For a real number a, both eigenvalues of*

$$B(a) = \begin{pmatrix} a & a^2 \\ a - 1 & a^2 \end{pmatrix}$$

lie strictly inside the unit disk if and only if $-1 < a < 1$. With $a = 1 - H/4$, this is exactly $0 < H < 8$.

Proof. The characteristic polynomial is

$$r^2 - a(1 + a)r + a^2.$$

For $-1 < a < 1$ and $a \neq 0$, its discriminant is $a^2(a - 1)(a + 3) < 0$. Its conjugate roots therefore have modulus $|a| < 1$. At $a = 0$,

$$B(0) = \begin{pmatrix} 0 & 0 \\ -1 & 0 \end{pmatrix}, \qquad B(0)^2 = 0.$$

This nonzero nilpotent matrix is defective but stable. If $|a| \geq 1$, the product of the eigenvalues is $a^2 \geq 1$, so both cannot have modulus below one. At $a = -1$ the eigenvalues are $\pm i$; at $a = 1$ both are 1. ∎

Lemma 2 (Collision-safe metric)**.** *On $-1 < a < 1$, set*

$$R(a) = 2a^2 + a + 1, \qquad D(a) = (1 - a^2)R(a)$$

and define a symmetric matrix $Q(a)$ by

$$\begin{aligned} Q_{11}(a) &= \frac{2 + 3a^2 - a^3 - a^4 - a^5}{D(a)}, \\ Q_{12}(a) &= \frac{a^2(a^3 + a^2 + 2a - 1)}{D(a)}, \\ Q_{22}(a) &= \frac{a^6 + 2a^4 - 2a^3 + 1}{(1 - a)D(a)}. \end{aligned} \tag{19}$$

Then $Q(a) > 0$, $Q(a) \geq I$, and

$$Q(a) - B(a)^* Q(a) B(a) = I, \qquad Q(0) = \mathrm{diag}(2, 1). \tag{20}$$

Rational functional calculus defines a positive metric $\mathcal{Q}_p$ on $N_pM \oplus N_pM$ with

$$I \leq \mathcal{Q}_p \leq M_Q I, \qquad B_p^* \mathcal{Q}_p B_p = \mathcal{Q}_p - I \tag{21}$$

for one finite $M_Q > 1$. The matrix function Q is analytic on its spectral domain; the bundle metric $\mathcal{Q}_p$ inherits the finite regularity of H_p, at least C^3 under the stated C^5 data. In particular all derivatives used below are uniformly bounded.

Proof. Every denominator is nonzero because

$$R(a) = 2(a + 1/4)^2 + 7/8 > 0.$$

Direct polynomial multiplication verifies (20). Positivity can be checked without presupposing stability:

$$Q_{22}(a) = \frac{(a^3 - 1)^2 + 2a^4}{(1 - a)D(a)} > 0,$$

$$\det Q(a) = \frac{a^4 + (a^2 - a)^2 + 2(1 - a)}{(1 - a)^3 (1 + a)^2 R(a)} > 0.$$

The two conditions imply positive definiteness. The Lyapunov identity then implies $Q(a) \geq I$.

In (19), replace a by A_p and scalar constants by multiples of the identity. The denominator operators $I - A_p$, $I - A_p^2$, and $I + A_p + 2A_p^2$ are uniformly invertible on the fixed compact spectral interval. Orthogonal diagonalization at a single p proves positivity and (21), because all blocks commute. No eigenbasis is required to vary with p. In local

orthonormal bundle frames, matrix inversion and multiplication preserve the available finite differentiability. Compactness gives the uniform derivative bounds. Thus eigenvalue crossings, repeated eigenvalues, and the defective $A_p = 0$ block require no additional hypothesis. ∎

The identity gives the uniform frozen-power bound

$$\|B_p^j\| \le \sqrt{M_Q}(1 - 1/M_Q)^{j/2}, \qquad j \ge 0.$$

A common Euclidean one-step contraction and stability under arbitrary switching are not being asserted. The actual moving base is controlled as follows.

There is $\rho_* < 1$ with $|B_p z|_{\mathcal{Q}_p} \le \rho_* |z|_{\mathcal{Q}_p}$. Since $p_+ - p = O(h)$, the change of Q under normal transport is $O(h)$. Fix ρ with $\rho_* < \rho < 1$. For small enough h, (15) gives

$$|z_+|_{\mathcal{Q}_{p_+}} \le \rho |z|_{\mathcal{Q}_p} + C_{R,D} h^2. \tag{22}$$

This contracts the two-history phase; an estimate on n alone would not suffice.

Iterating while the cap assumptions hold, with $j = k - K$, gives

$$|z_k| \le C\rho^j |z_K| + C_{R,D} h_k^2 \le C\theta h_K \rho^j + C_{R,D} h_k^2. \tag{23}$$

Indeed, the initial Q-norm is at most $\sqrt{2M_Q}\theta h_K$, and

$$\sum_{i=K}^{k-1} \frac{\rho^{k-1-i}}{i} \le \frac{1}{k} \sum_{\ell=0}^{\infty} \rho^\ell (\ell + 2) \le \frac{C_\rho}{k}.$$

Here $k/i = 1 + (k-i)/i \le \ell + 2$ for $\ell = k - 1 - i$ and $i \ge 1$. Constants cover $k = K$ as well. In particular,

$$\sup_{j \ge 0} \frac{h_K}{h_{K+j}} \rho^j \le \sup_{j \ge 0} \sqrt{1 + j}\, \rho^j < \infty.$$

Choose D larger than four times the resulting coefficient of θ in the bound for $(|n|+|u|)/h$, plus 4. Here θ may be any one fixed number in $(0, 1]$, chosen once for the theorem. Increasing K later makes the forcing contribution $C_{R,D} h_k$ smaller than 1 uniformly. Thus (23) improves the cap $|n| + |u| \le Dh$ to a strict interior cap, for example $(D/2)h$.

Equations (17), (23), and $|z| \le Dh$ imply

$$|r_k^p| \le C h_k^2, \qquad |r_k^v| \le C h_k^2 + t_k, \qquad t_k = C\theta h_K \rho^{k-K}. \tag{24}$$

These constants may depend on the prescribed tangential cap, but not on the admitted history. In particular, $\sum_{k \ge K} t_k \le C\theta h_K$. The entry transient remains explicit; the normal variables have not been assumed to be $O(1/k)$ immediately upon entry.

2.4 A strict slow-clock Lyapunov function

Let

$$G = \max_M |g|, \qquad Q_{\max} = \max_M q, \qquad L_q = \max_M |\operatorname{Hess}_M q|.$$

Choose a fixed $\varepsilon > 0$ with

$$\varepsilon \le 1, \qquad \varepsilon(L_q + 9/2) \le 3/4.$$

On TM, define

$$\mathcal{E}(p, v) = \tfrac{1}{2}|v|^2 + bq(p) + \varepsilon\langle g(p), v\rangle.$$

Use the intrinsic vector field

$$p' = v, \qquad \nabla_t v = -\gamma v - bg(p)$$

only as an auxiliary differentiation device. Its exact derivative of $\mathcal{E}$ is

$$D\mathcal{E} = -\gamma|v|^2 + \varepsilon \operatorname{Hess}_M q[v, v] - \varepsilon\gamma\langle g, v\rangle - \varepsilon b|g|^2.$$

Since

$$\gamma|g|\,|v| \le \frac{b}{2}|g|^2 + \frac{\gamma^2}{2b}|v|^2, \qquad \frac{\gamma^2}{2b} = \frac{9}{2},$$

we obtain

$$D\mathcal{E} \le -\tfrac{3}{4}|v|^2 - \tfrac{\varepsilon}{8}|g|^2 \le -c\big(|v|^2 + |g|^2\big), \qquad c = \min(3/4, \varepsilon/8) > 0. \tag{25}$$

The cross term supplies dissipation of the restricted-norm gradient as well as velocity. No isolated-critical-point or convexity assumption on $q|_M$ is needed.

We now prove an inequality for the actual finite-history iterates. On every fixed velocity cap, the horizontal and vertical first derivatives satisfy

$$|D_p\mathcal{E}| + |D_v\mathcal{E}| \le C(|v| + |g|),$$

and the second derivatives are uniformly bounded. Write $A_0 = |v| + |g|$. Apply second-order Taylor expansion in the exponential and parallel-transport coordinates of (16)–(17). The leading increment is $h(v, -\gamma v - bg)$, with residual increments r^p, r^v. Equations (24) and (25) give

$$\begin{aligned}\mathcal{E}(p_+, v_+) - \mathcal{E}(p, v) \le\ & -ch(|v|^2 + |g|^2) + Ch^2A_0^2 + Ch^2A_0 \\ & + Ct_k + C(h^4 + t_k^2).\end{aligned}$$

Terms multiplying t_k are uniformly bounded on the fixed cap; for large K, $t_k \le 1$ and $t_k^2 \le t_k$. For small h, half of the negative term absorbs the first positive term. Young's inequality gives

$$Ch^2A_0 \le \frac{c}{4}h(|v|^2 + |g|^2) + C'h^3.$$

Also $h^4 \le h^3$. Decreasing the dissipation constant if needed, we conclude that

$$\begin{aligned}\mathcal{E}(p_{k+1}, v_{k+1}) - \mathcal{E}(p_k, v_k) \le & -c_1 h_k \big(|v_k|^2 + |g(p_k)|^2\big) \\ & + C h_k^3 + C t_k,\end{aligned} \tag{26}$$

for a fixed $c_1 > 0$. Both positive error sequences are summable:

$$\sum_{k=K}^{\infty} h_k^3 \le C h_K, \qquad \sum_{k=K}^{\infty} t_k \le C\theta h_K.$$

This is an absorbed, summable discrete energy estimate. No bare $O(h^2)$ additive energy error is being summed.

2.5 Closing the capture bootstrap

Fix the prescribed B_0. The entry energy satisfies

$$\mathcal{E}(p_K, v_K) \le \mathcal{E}_0 := \tfrac{1}{2} B_0^2 + b Q_{\max} + \varepsilon G B_0.$$

Since $q \ge 0$, completion of a square gives, for every p, v,

$$\mathcal{E}(p, v) \ge \tfrac{1}{4}|v|^2 - \varepsilon^2 G^2. \tag{27}$$

Choose finite $R > B_0 + 2$ so large that

$$\tfrac{1}{4}(R-1)^2 - \varepsilon^2 G^2 > \mathcal{E}_0 + 3.$$

Choose D as in Section 2.3. Put

$$\beta = \sup_{p \in M} \|B_p\|, \qquad D_{\text{aux}} = \max\{D + 1, 2\beta D + 2\},$$

where the operator norm is the Euclidean product norm. This local notation β is independent of the source algorithm's parameter notation above. All one-step, contraction, Taylor, and energy constants are now fixed on the working caps R, D and, when necessary, the larger finite caps $R + 1, D_{\text{aux}}$. Choose the original integer $K_0(B_0)$ sufficiently large that:

1. all one-step constructions are valid on these caps;
2. all points within the estimated $O(h)$ distances of M stay in the chosen domain and tubes;
3. the strict normal-cap improvement in Section 2.3 holds;
4. the summed positive errors in (26) from any $K \ge K_0$ are less than 1;

5. the raw next-step bounds below place the next velocity in the $R+1$ cap and the next scaled normal state in the D_{aux} cap.

Here are input-only justifications of the last condition. Section 2.2 constructs the predecessor, predictor, next point, and next projection in the larger tube before any next-state phase cap is assumed. Isometry of normal transport and (15) give

$$|z_+|_2 \le \beta|z|_2 + C_{R,D}h^2.$$

The sum norm is at most $\sqrt{2}$ times the Euclidean product norm, and $h/h_+ \le \sqrt{2}$ for $k \ge 1$. Thus

$$\frac{|n_+| + |u_+|}{h_+} \le 2\beta D + 2C_{R,D}h < D_{\mathrm{aux}}$$

for large K_0. Likewise, (17), $|z| \le Dh$, and isometry of tangent transport imply

$$\big||v_+| - |v|\big| \le C_{R,D}h < 1,$$

so $|v_+| < R+1$. Neither raw estimate uses a strict next-state working cap. The requirements are finitely many smallness conditions on h_K with constants fixed before K, so such an integer K_0 exists. This choice preserves the original clock.

The entry test gives $|v_K| \le B_0 < R-1$ and

$$|n_K| + |u_K| \le 2\theta h_K < Dh_K.$$

It also gives

$$|x_{K-1} - p_K| \le |n_K| + |u_K| + h_K|v_K| \le (2\theta + B_0)h_K,$$

placing the predecessor in the domain without any additional hypothesis.

Induct up to a possible first violation of the tube or working phase caps. Suppose the working caps hold through index k. The input-only one-step construction and raw estimates first define the next point and place it in the larger tube and auxiliary phase caps. Apply (22) to this just-constructed step and iterate through index $k+1$. This requires only the current working caps and the $O(h_k)$ displacement of the already-defined next base; it does not assume the next normal state is in the working cap. Equation (23) therefore improves that next state to

$$|n_{k+1}| + |u_{k+1}| \le (D/2)h_{k+1}.$$

The raw bound $|v_{k+1}| < R+1$ supplies the tangent-bundle neighborhood for the energy Taylor estimate on the same step. Its errors use the current normal bound and historical transient, not a next-state working-cap assumption. Consequently (26) holds through this step. Summing through index k gives

$$\mathcal{E}(p_{k+1}, v_{k+1}) \le \mathcal{E}_0 + 1.$$

By (27) and the choice of R, this improves the tangent bound to $|v_{k+1}| < R-1$. The normal distance is at most $(D/2)h_{k+1}$, and the predictor normal distance on the constructed step is at most $C(Dh_k + h_k^2)$, so the tube bounds improve as well for large K_0. A first violation is impossible. This proves all-time existence and control of the actual suffix, together with (23).

Since $\mathcal{E}$ is bounded below, summing (26) gives

$$\sum_{k=K}^{\infty} h_k\big(|v_k|^2 + |g(p_k)|^2\big) < \infty. \tag{28}$$

The argument works for each prescribed finite B_0. Increasing B_0 changes the phase cap and how late the original entry index must be. It does not restrict p_K to a particular component, well, or basin on M.

2.6 Vanishing phase and critical-set capture

The phase recurrences imply

$$\big||v_{k+1}| - |v_k|\big| \le Ch_k + t_k,$$

and tangent transport τ_k from p_k to p_{k+1} gives

$$|g(p_{k+1}) - \tau_k g(p_k)| \le C\,\mathrm{dist}_M(p_{k+1}, p_k) \le Ch_k.$$

The ratio estimate of Section 2.3 bounds $t_k \le Ch_k$ uniformly for $k \ge K$. Hence, for $a_k = |v_k|^2 + |g(p_k)|^2$,

$$|a_{k+1} - a_k| \le Ch_k. \tag{29}$$

Together with $a_k \ge 0$, $\sum h_k a_k < \infty$, $h_k \to 0$, and $\sum h_k = \infty$, this implies $a_k \to 0$.

For a direct discrete proof, suppose $a_k \ge \delta > 0$ at arbitrarily large indices. Equation (29) keeps $a_j \ge \delta/2$ on the ensuing slow-time interval until $\sum h_j$ first reaches $\delta/(2C)$, apart from an endpoint error that vanishes as $k \to \infty$. Each such interval contributes a uniformly positive amount to $\sum h_j a_j$. Infinitely many disjoint intervals may be selected, unless a stays bounded below forever; the latter also contradicts summability because $\sum h_j = \infty$. Both possibilities are impossible.

Thus $v_k \to 0$ and $g(p_k) \to 0$. Compactness makes $\mathrm{Crit}_M q$ nonempty and compact. It also implies

$$\mathrm{dist}(p_k, \mathrm{Crit}_M q) \longrightarrow 0 :$$

otherwise a subsequence staying a positive distance away would have a convergent subsequence with zero gradient and positive distance from the critical set. Since $n_k \to 0$, the same distance conclusion holds for x_k.

The suffix stays in a bounded tube, so its omega-limit set is nonempty and compact. All cluster points belong to $\mathrm{Crit}_M\, q$. Finally,

$$d_k = h_k v_k + u_k \longrightarrow 0.$$

The gradients at x_k and x_{k-1} differ by $O(|d_k|)$ in the controlled neighborhood. Equation (2) therefore gives $y_k - x_k \to 0$. Bounded sequences whose pointwise difference tends to zero have identical omega-limit sets, proving the predictor assertion.

2.7 Curvature correction and normal tracking

Equation (23) already implies $|z_k| = O(h_k^2)$ as $k \to \infty$, because the exponential transient is eventually dominated by every power of k. All estimates in this section are uniform on the controlled velocity cap; their constants may depend on the admitted entry.

At a sufficiently late step write $h = h_k$, $p = p_k$, $v = v_k$, $A = A_p$, $H = H_p$, and $S = \mathrm{II}_p(v, v)$. Since $n, u = O(h^2)$, Taylor expansion at p and (14) give

$$\nabla F(x) - \nabla F(x_{\mathrm{prev}}) = Hu + \tfrac{1}{2}h^2 HS + O(h^3). \tag{30}$$

Explicitly, $x_{\mathrm{prev}} = p + n - u - hv$, so

$$\begin{aligned}\nabla F(x_{\mathrm{prev}}) &= H(n-u) + \tfrac{1}{2}h^2 D^3F(p)[v,v] + O(h^3)\\ &= H(n-u) - \tfrac{1}{2}h^2 HS + O(h^3),\end{aligned}$$

whereas $\nabla F(x) = Hn + O(h^4)$. This verifies the curvature sign in (30).

The predictor has tangent coordinate $hv - \gamma h^2 v + O(h^3)$ at p and ambient normal coordinate

$$n + Au - \tfrac{b}{2}h^2 HS + O(h^3).$$

Its normal displacement relative to its own projected base, represented back at p to the needed order, is therefore

$$n + Au - \tfrac{1}{2}h^2(I + bH)S + O(h^3). \tag{31}$$

The normal part of $x_+ - p$ at the old base equals

$$An + A^2 u + \tfrac{b^2}{2}h^2 H^2 S - bh^2 N_p p + O(h^3).$$

By (12) and $\log_p p_+ = hv + O(h^2)$, the normal part of $p_+ - p$ is $h^2 S/2 + O(h^3)$. Subtracting yields

$$\bar{n}_+ = An + A^2 u - \tfrac{1}{2}h^2 A(I + bH)S - bh^2 N_p p + O(h^3), \tag{32}$$

using $I - b^2H^2 = A(I + bH)$.

The normal projection at the *new* base of the chord $p_+ - p$ is $-h^2 S/2 + O(h^3)$, represented at p. This sign differs from its old-base normal projection. Thus

$$\bar{u}_+ = \bar{n}_+ - n - \tfrac{1}{2}h^2 S + O(h^3). \tag{33}$$

Transport errors here are $O(h^3)$ because the normal vectors have size $O(h^2)$ and the base moves by $O(h)$. These formulas retain curvature from both the gradient-difference term and the new-base history convention.

Equations (32)–(33) take the form

$$\bar{z}_+ = B_p z + h^2 f(p, v) + O(h^3),$$

where

$$f_n = -\tfrac{1}{2}A(I + bH)S - bN_p p, \qquad f_u = f_n - \tfrac{1}{2}S.$$

Solve the algebraic frozen equation $\zeta = B_p\zeta + f$. Its unique solution is

$$\begin{aligned}\zeta_u &= -\tfrac{1}{2}S,\\ \zeta_n &= -H^{-1}N_p p - b^{-1}H^{-1}AS\\ &= -H^{-1}N_p p - (4H^{-1} - I)\mathrm{II}_p(v, v).\end{aligned}$$

For verification, substitute $\zeta_u = -S/2$ into the first row. Its curvature coefficient becomes

$$-\tfrac{1}{2}\big[A^2 + A(I + bH)\big]S = -AS,$$

because $A + I + bH = 2I$. Hence

$$(I - A)\zeta_n = -AS - bN_p p,$$

which gives the displayed formula since $I - A = bH$. The second row reduces to $\zeta_u = -S/2$. Invertibility also follows from the strict stability of B_p.

The algebraic target is not assumed invariant. It has sufficient regularity on the bounded tangent-bundle region. The established phase estimates and $|z| = O(h^2)$ give

$$\mathrm{dist}_M(p_+, p) = O(h), \qquad \bar{v}_+ - v = O(h), \qquad h_+^2 - h^2 = O(h^4).$$

Consequently,

$$\overline{h_+^2\zeta(p_+, v_+)} - h^2\zeta(p, v) = O(h^3). \tag{34}$$

Set $e_k = z_k - h_k^2\zeta(p_k, v_k)$. Equations (32)–(34), the Q-norm construction, and a possibly later fixed index J imply

$$|e_{k+1}|_{Q_{p_{k+1}}} \le \rho|e_k|_{Q_{p_k}} + Ch_k^3, \qquad k \ge J,$$

for a fixed $\rho < 1$, enlarged from its earlier value if necessary. Iteration gives $|e_k| = O(h_k^3)$: the initial exponential term is eventually $O(h_k^3)$, and the forcing convolution is of this order because

$$(k/i)^{3/2} \le (k - i + 1)^{3/2}$$

and a geometric sequence multiplied by this polynomial is summable. This proves both expansions in (8).

Finally $v_k \to 0$, and H_p^{-1} and II_p are uniformly bounded, so

$$\begin{aligned} kn_k + H_{p_k}^{-1} N_{p_k} p_k &= -(4H_{p_k}^{-1} - I)\mathrm{II}_{p_k}(v_k, v_k) + O(k^{-1/2}) \\ &\longrightarrow 0. \end{aligned}$$

The curvature-dependent target is needed in the intermediate bounded-velocity regime, although its curvature contribution vanishes in the eventual critical-capture limit. This completes the proof of Theorem 1.

2.8 A phase-sensitive interface for selection

Lemma 3 (Phase-sensitive actual increments)**.** *On a late suffix of every captured trajectory, let τ_k be tangent parallel transport from p_k to p_{k+1}. Then*

$$\begin{aligned} \log_{p_k} p_{k+1} &= h_k v_k + r_k^p, \\ \tau_k^{-1} v_{k+1} &= v_k - h_k(\gamma v_k + bg(p_k)) + r_k^v, \\ |r_k^p| &\le C\{h_k^2(|v_k| + |g(p_k)|) + h_k^4\}, \\ |r_k^v| &\le C\{h_k^2(|v_k| + |g(p_k)|) + h_k^3\}. \end{aligned} \tag{35}$$

The constants may depend on the captured history, but the phase-independent velocity error retains order h^3 over the whole interval $L < 8$.

Proof. The calculation uses only capture and the literal two-line map. Write $V = |v|$, $G = |g|$, and use $n, u = O(h^2)$ from capture. For small tangent a and normal η at p,

$$\mathrm{dist}_M(p, \pi(p + a + \eta)) \le C|a|,$$

because $\pi(p + \eta) = p$. The predecessor has tangent displacement $-hv$ and normal displacement $O(h^2)$, hence

$$|P_p \nabla F(x)| \le Ch^4, \qquad |P_p \nabla F(x_{\mathrm{prev}})| \le C(h^3 V + h^4).$$

The predictor tangent coordinate is $hv - \gamma h^2 v + O(h^3 V + h^4)$; its normal coordinate is $O(h^2)$ because A is bounded. The projected predictor base moves by $O(hV + h^4)$, so its tangent gradient is also $O(h^3 V + h^4)$. Applying the actual corrector therefore gives

$$P_p(x_+ - x) = hv - h^2(\gamma v + bg) + O(h^3(V + G) + h^4).$$

The resulting base displacement satisfies $\ell = \mathrm{dist}_M(p, p_+) \le C(hV + h^2 G + h^4)$. Tangent projection versus tangent transport adds $O(\ell |N_p(x_+ - x)| + \ell^2 |P_p(x_+ - x)|)$, where the normal increment is $O(h^2)$. Projection of the normal position contributes the same order. Finally $h/h_+ = 1 + O(h^2)$. Dividing the increment by h_+ and using bounded V, G yields (35). We also obtain

$$\mathrm{dist}_M(p_{k+1}, p_k) \le C\{h_k |v_k| + h_k^2 |g(p_k)| + h_k^4\}. \tag{36}$$

Every occurrence of A in this calculation is a bounded polynomial multiplier. No positivity or inverse of A enters. ∎

The moving metric has one further reusable consequence. If a normal-pair error satisfies

$$\bar{e}_{k+1} = B_{p_k} e_k + r_k, \qquad |r_k| \le C h_k^m,$$

and the already constructed base step is $O(h_k)$, then

$$|e_k| \le C\rho^{k-J}|e_J| + C h_k^m. \tag{37}$$

This also permits an $O(h_k)$ perturbation of B_{p_k}. It follows from Lemma 2 and

$$\frac{h_i^m}{h_k^m} = \left(\frac{k}{i}\right)^{m/2} \le (k-i+1)^{m/2}, \qquad J \le i < k.$$

This is forward filtering of a stable pair. Neither B_p^{-1} nor arbitrary-switching stability is required.

The remaining proofs use this phase lemma, the retained current-base history, and the fast filter. Their tangent hyperbolicity and KL estimates involve only the intrinsic Hessian of q, never an inverse or positivity of A_p. Every normal contraction below is supplied by Lemma 2; all constants are selected on the fixed spectral band in (1).

3 Morse and Morse–Bott finite actual travel

A connected component S of $\mathrm{Crit}_M q$ is called Morse–Bott here if it is a compact embedded critical submanifold without boundary and

$$\ker \mathrm{Hess}_M q(s) = T_s S, \qquad s \in S. \tag{38}$$

It is enough initially that this embedded submanifold be C^1: Section 3.1 identifies it with a C^4 implicit graph in fixed C^5 charts of M. The condition says that the Hessian is nonsingular on the orthogonal complement of $T_s S$ inside $T_s M$. Different critical components may have different dimensions.

Theorem 2 (Morse–Bott finite actual travel)**.** *Assume* (1)*, and suppose every connected component of* $\mathrm{Crit}_M q$ *is compact and satisfies* (38)*. There are only finitely many such components. Every history admitted by Theorem 1 has a critical component S and a point $p_* \in S$ such that $x_k \to p_*$ and $y_k \to p_*$. The predecessor is in the domain of F for this threshold. For the admitted history,*

$$\begin{aligned} \mathrm{dist}_M(p_k, S) + |v_k| + |g(p_k)| &= O(k^{-1}), \\ \mathrm{dist}(x_k, S) + \mathrm{dist}(y_k, S) &= O(k^{-1}). \end{aligned} \tag{39}$$

The bound on v_k concerns its entire *tangent vector, including directions tangent to S. The literal increments satisfy*

$$|x_k - x_{k-1}| + |y_k - x_k| + |y_{k+1} - y_k| = O(k^{-3/2}). \tag{40}$$

Both trajectories have finite total travel, with tails

$$\sum_{j=k+1}^{\infty} |x_j - x_{j-1}| + \sum_{j=k}^{\infty} |y_{j+1} - y_j| = O(k^{-1/2}). \tag{41}$$

The corresponding distance to the selected point obeys

$$\mathrm{dist}_M(p_k, p_*) + |x_k - p_*| + |y_k - p_*| = O(k^{-1/2}). \tag{42}$$

On a sufficiently late suffix, let s_k be the intrinsic nearest-point projection of p_k onto S. Then

$$s_k \to p_*, \qquad \mathrm{dist}_S(s_k, p_*) = O(k^{-1/2}). \tag{43}$$

All asymptotic constants and the late suffix may depend on the admitted history. The entry constants are those constructed in Theorem 1; the original clock is unchanged. No entry restriction to a secondary basin or extra smallness of B_0 is imposed.

Theorem 3 (Localized form)**.** *For a particular admitted trajectory, the global Morse–Bott condition may be replaced by the hypothesis that its omega-limit set meets one compact connected Morse–Bott critical submanifold S which is a connected component of* $\mathrm{Crit}_M q$*. Then the trajectory conclusions* (39)–(43) *of Theorem 2 hold for this trajectory and this component.*

The $O(k^{-1})$ estimate in (39) is a transverse estimate, not a distance-to-p_* estimate for a positive-dimensional S. Section 4 identifies its coefficient and proves attained sharpness. When S is a point, the fiber estimate below with no center position yields the stronger $O(k^{-1})$ pointwise rate. If q is constant on a connected component of M, then S may be that entire component: the transverse position bundle has rank zero, but all tangent velocities remain damped stable modes.

Corollary 3.1 (Morse pointwise bounds and isolated-point localization)**.** *If $q|_M$ is Morse, or a captured trajectory has one nondegenerate critical cluster point p_*, then that point is the unique limit, both actual paths have finite travel, and*

$$\mathrm{dist}_M(p_k, p_*) + |v_k| + |x_k - p_*| + |y_k - p_*| = O(k^{-1}).$$

The increment, offset and travel-tail bounds of Theorem 2 still hold. Merely having an isolated critical cluster point, possibly degenerate, gives point convergence of p_k, x_k, y_k, without a finite-travel or rate conclusion.

3.1 Critical components and their finite regularity

Lemma 4. *Every compact Morse–Bott critical component S is locally a C^4 submanifold and is isolated within* $\mathrm{Crit}_M q$. *If all critical components have this property, there are finitely many. The omega-limit set of an admitted history is connected and lies in one such component. In the localized form, meeting S suffices to put the entire omega-limit set in S.*

Proof. Fix $s \in S$. Start with a *fixed C^5 chart of M*, centered at s and with orthonormal derivative there. Split its coordinate space linearly as

$$T_sM = K \oplus V, \qquad K = T_sS, \quad V = K^\perp \cap T_sM.$$

The coordinate representation $\widetilde{g}(a, z)$ of g is C^4. At the critical point its derivative is the Hessian endomorphism $Q_s = \mathrm{Hess}_M q(s)$. By self-adjointness and (38), the derivative with respect to z of the fixed V-component

$$(a, z) \longmapsto \Pi_V \widetilde{g}(a, z)$$

is the invertible map $Q_s|_V$. The C^4 implicit function theorem therefore gives a unique C^4 graph $z = \psi(a)$ solving these V-component equations near s.

Since S is initially an embedded C^1 submanifold with tangent K, its projection onto K is a local C^1 diffeomorphism. Thus S is a C^1 graph $z = \chi(a)$ on a smaller neighborhood. Every point of this graph is critical, so it solves the V-component equations. Uniqueness of the implicit graph gives $\chi = \psi$. This identifies the pre-existing S with a C^4 graph in the fixed chart. Every full critical point near s also lies on this graph, and every point on it belongs to S. Hence the critical set there is exactly S.

This argument neither assumes a previously smooth chart adapted to S nor uses the desired regularity to obtain that chart. It supplies the finite regularity needed later: the tubular and transport maps have at least the bounded second derivatives required for their Taylor estimates, and the bundle projections and metrics have the bounded first derivatives required for slow variation. No analytic or C^∞ hypothesis is added. Below, smooth dependence means only these finite orders.

A finite cover of compact S by the preceding neighborhoods isolates S from all other critical points. If every component is compact and Morse–Bott, the locally isolating neighborhoods cover the compact set $\mathrm{Crit}_M q$. A finite subcover shows that only finitely many components occur.

Theorem 1 gives a nonempty compact omega-limit set $\Omega \subset \mathrm{Crit}_M q$ and vanishing actual steps. Such an omega-limit set is connected. Indeed, a separation $\Omega = A \cup B$ into nonempty disjoint compact sets would give small neighborhoods of A and B at positive distance. All sufficiently late terms must lie in their union; otherwise a subsequence outside has a cluster point outside Ω. Vanishing steps prevent passage between the two

late neighborhoods, contradicting cluster points in both. Since $n_k \to 0$, the p sequence has the same Ω.

Connectedness now places Ω in one of the finitely many isolated components. In the localized case, S is relatively open and relatively closed in $\mathrm{Crit}_M\, q$, so the same argument applies as soon as Ω meets S. Compactness gives

$$\mathrm{dist}_M(p_k, S) \to 0, \qquad |v_k| \to 0. \tag{44}$$

This establishes neither one-point convergence within S nor finite length. At this stage Ω may still be a nontrivial subset of S. ∎

The same connectedness argument shows that an isolated critical cluster point is the entire cluster set. If it is nondegenerate, its singleton is a Morse–Bott component. If q is Morse on compact M, all critical points are isolated, and compactness makes their number finite. This proves the localization assertions of Corollary 3.1; its stronger point bound follows below.

3.2 The phase bundle retains all tangent velocities

Write $m = \dim M$ and $d_S = \dim S$. For $s \in S$, set

$$K_s = T_s S, \qquad V_s = K_s^{\perp} \cap T_s M.$$

Use the intrinsic normal exponential tube of S in M. By (44), on a sufficiently late suffix there are unique

$$s_k \in S, \qquad \eta_k \in V_{s_k}, \qquad p_k = \exp^M_{s_k}(\eta_k).$$

Parallel-transport v_k from p_k back to s_k along this short normal geodesic, and denote the result by w_k. The fiber variable and bundle are

$$Z_k = (\eta_k, w_k) \in \mathcal{E}_{s_k}, \qquad \mathcal{E}_s = V_s \oplus T_s M. \tag{45}$$

This bundle has rank $2m - d_S$ and need not be trivial. Compactness and tubular regularity give uniform equivalence and bounds

$$|Z_k| \asymp \mathrm{dist}_M(p_k, S) + |v_k|, \qquad |g(p_k)| \le C|\eta_k|, \qquad Z_k \to 0. \tag{46}$$

In particular w_k includes its K_s component. Removing velocities tangent to S together with equilibrium-position centers would lose the damping needed for the result.

The Hessian $Q_s = \mathrm{Hess}_M\, q(s)$ kills K_s and, by self-adjointness, preserves V_s. Define the endomorphism

$$L_s(\eta, w) = \big(P_{V_s} w, -bQ_s\eta - \gamma w\big) \quad \text{on } \mathcal{E}_s. \tag{47}$$

For each nonzero eigenvalue λ of $Q_s|_{V_s}$, the corresponding two characteristic roots solve

$$r^2 + \gamma r + b\lambda = 0. \tag{48}$$

For $\lambda > 0$, both roots have negative real part. For $\lambda < 0$, one root is positive and the other negative. Each of the d_S velocity directions tangent to S contributes the stable root $-\gamma$. Thus L_s is hyperbolic on $\mathcal{E}_s$, even at transverse saddles and maxima. Its equilibrium-position centers have been excluded, but its stable tangent-velocity modes are retained. Compactness and nondegeneracy give a uniform gap from the imaginary axis. No eigenbasis is required.

For close $s, s_+ \in S$, let

$$T_{s_+,s} : \mathcal{E}_s \longrightarrow \mathcal{E}_{s_+}$$

be isometric transport, using the normal connection of $S \subset M$ on the V component and the Levi-Civita connection of M along the short geodesic in S on the TM component. These definitions work on a nontrivial bundle. When $d_S = 0$, the connected component S is one point, the base is fixed, and this transport is the identity.

Lemma 5 (Finite-step fiber and base estimates)**.** *Put $T_k = T_{s_{k+1},s_k}$. On a sufficiently late original suffix,*

$$\begin{aligned} T_k^{-1} Z_{k+1} &= (I + h_k L_{s_k}) Z_k + R_k, \\ |R_k| &\le C\big[h_k |Z_k|^2 + h_k^2 |Z_k| + h_k^3\big]. \end{aligned} \tag{49}$$

Independently of finite length or convergence of (s_k),

$$d_k^S := \operatorname{dist}_S(s_{k+1}, s_k) \le C\big[h_k |Z_k| + h_k^4\big] = o(h_k). \tag{50}$$

Proof. Transform the literal finite-step estimates (35) through the tubular phase coordinates above. The intrinsic field naming their first-order term is

$$p' = v, \qquad \nabla_t v = -\gamma v - b g(p).$$

At $(s, \eta, w) = (s, 0, 0)$, the derivative of its base component is $P_{K_s} w$. The derivative of its fiber component is exactly

$$(\eta, w) \longmapsto (P_{V_s} w, -b Q_s \eta - \gamma w) = L_s(\eta, w).$$

Terms caused by the changing tubular splitting, normal transport, and connection vanish to second order in (η, w). They contribute $O(h|Z|^2)$. The transformed discrete remainder and the finite-step coordinate Taylor remainder are $O(h^2|Z| + h^3)$: use $|g| \le C|\eta|$ in the raw estimates and a step bounded by $C(h|Z| + h^3)$. Uniformly bounded second derivatives on finitely many restricted charts justify these Taylor bounds. This proves (49) directly from the map estimates; no differential-equation approximation or invariant-manifold theorem is invoked.

The tubular projection onto S is uniformly Lipschitz locally. Apply it to (36), use (46), and absorb $h_k^2|\eta_k|$ into $h_k|Z_k|$. This proves (50). Finally $Z_k \to 0$ implies $d_k^S / h_k \to 0$. Its sum is not assumed finite at this point. ∎

3.3 A dichotomy on the moving bundle

3.3.1 Spectral subbundles and Lyapunov metrics

Let P_s^- and P_s^+ be the stable and unstable spectral projections of L_s, and let $\mathcal{E}_s^-$ and $\mathcal{E}_s^+$ be their ranges. The superscripts refer to spectral sign, while the subscript denotes the base point. Separating contours for the stable and unstable spectral regions define the projections by resolvent integrals in local bundle frames. Uniqueness makes these local definitions agree on overlaps. They have the bounded derivatives needed below. Eigenvalues inside a stable block may collide or be nonsemisimple; this requires no smooth eigenvectors.

On the stable and unstable subbundles, respectively, define positive quadratic forms

$$\begin{aligned} \mathcal{Q}_s^-(z,z) &= \int_0^\infty |\exp(tL_s)z|^2\,dt, \qquad z \in \mathcal{E}_s^-, \\ \mathcal{Q}_s^+(z,z) &= \int_0^\infty |\exp(-tL_s)z|^2\,dt, \quad z \in \mathcal{E}_s^+. \end{aligned} \tag{51}$$

Compactness and uniform spectral separation imply convergence, uniform norm equivalence, and bounded first derivatives. To justify uniform exponential bounds under the integrals, use finite-dimensional resolvent contours on a finite cover of S and decrease the spectral rate if needed. Possible polynomial factors from Jordan blocks are absorbed by this decrease. Differentiating with respect to the base introduces only integrable exponential-polynomial factors.

Integration of the derivative in t gives

$$2\mathcal{Q}_s^-(L_s z, z) = -|z|^2, \qquad 2\mathcal{Q}_s^+(-L_s z, z) = -|z|^2.$$

Consequently there are uniform $a_0 > 0$ and $h_0 > 0$ such that, for $0 < h \le h_0$,

$$\begin{aligned} |(I + hL_s)z|_{\mathcal{Q}_s^-} &\le (1 - 2a_0 h)|z|_{\mathcal{Q}_s^-}, \quad z \in \mathcal{E}_s^-, \\ |(I + hL_s)^{-1}z|_{\mathcal{Q}_s^+} &\le (1 - 2a_0 h)|z|_{\mathcal{Q}_s^+}, \quad z \in \mathcal{E}_s^+. \end{aligned} \tag{52}$$

Choose h_0 also so that $I + hL_s$ is invertible and the displayed factors are positive. The stable estimate follows by expanding the square in the Lyapunov form. The unstable estimate follows by the uniform expansion $(I + hL_s)^{-1} = I - hL_s + O(h^2)$ in the unstable Lyapunov form.

3.3.2 Transport corrected to preserve spectral blocks

The isometry T_k need not preserve the spectral splitting. Define instead

$$D_k = \mathsf{P}_{s_{k+1}}^- T_k \mathsf{P}_{s_k}^- + \mathsf{P}_{s_{k+1}}^+ T_k \mathsf{P}_{s_k}^+. \tag{53}$$

Smooth dependence of the projections gives

$$|D_k - T_k| \le C d_k^S. \tag{54}$$

For sufficiently close bases this is an invertible map, by its small operator-norm distance from the isomorphism T_k. It maps each current spectral subspace bijectively onto the corresponding next one. The two subbundles have constant ranks on connected S.

In the varying Lyapunov norms,

$$\begin{aligned} |D_k z|_{\mathcal{Q}^\sigma_{s_{k+1}}} &\le (1 + C d_k^S)|z|_{\mathcal{Q}^\sigma_{s_k}}, \qquad z \in \mathcal{E}^\sigma_{s_k}, \\ |D_k^{-1} z_+|_{\mathcal{Q}^\sigma_{s_k}} &\le (1 + C d_k^S)|z_+|_{\mathcal{Q}^\sigma_{s_{k+1}}}, \qquad z_+ \in \mathcal{E}^\sigma_{s_{k+1}}, \end{aligned} \quad \sigma \in \{-, +\}. \tag{55}$$

For verification, take a compact local frame containing two close bases. Transport differs from the identity in that frame by $O(d_k^S)$, and both projection coefficients and metric coefficients vary by $O(d_k^S)$. Uniform positivity of the metrics yields the norm bounds; inversion gives the second one. A finite cover makes the constants global. There are no frame jumps in this intrinsic construction and no hypothesis on accumulated total variation.

Set

$$A_k = D_k(I + h_k L_{s_k}) : \mathcal{E}_{s_k} \longrightarrow \mathcal{E}_{s_{k+1}}. \tag{56}$$

These maps preserve the moving stable/unstable splitting. Because $d_k^S = o(h_k)$, (52)–(55) give, on a sufficiently late preliminary suffix,

$$\begin{aligned} |A_k z_-|_{\mathcal{Q}^-_{s_{k+1}}} &\le (1 - a_0 h_k)|z_-|_{\mathcal{Q}^-_{s_k}}, \\ |A_k^{-1} z_+|_{\mathcal{Q}^+_{s_k}} &\le (1 - a_0 h_k)|z_+|_{\mathcal{Q}^+_{s_{k+1}}}. \end{aligned} \tag{57}$$

For a late starting index J, put

$$t_k = \sum_{m=J}^{k-1} h_m.$$

Let $\Phi(k, l) : \mathcal{E}_{s_l} \to \mathcal{E}_{s_k}$ be the product of the A maps for $k > l$, the identity for $k = l$, and the product of their inverses for $k < l$. Multiplying (57) and using uniform norm equivalence yields fixed $C_A \ge 1$ and $a > 0$ such that

$$\begin{aligned} |\Phi(k, l)\mathsf{P}^-_{s_l}| &\le C_A e^{-a(t_k - t_l)}, \quad k \ge l \ge J, \\ |\Phi(k, l)\mathsf{P}^+_{s_l}| &\le C_A e^{-a(t_l - t_k)}, \quad J \le k \le l. \end{aligned} \tag{58}$$

This is a proved moving-bundle exponential dichotomy. Its premise is the slow variation $d_k^S = o(h_k)$, not an assertion that pointwise hyperbolic matrices remain stable under arbitrary switching. The base may still have infinite total travel at this stage.

3.3.3 An exact small perturbation along the actual trajectory

Rewrite (49) using (56):

$$Z_{k+1} = A_k Z_k + T_k R_k + (T_k - D_k)(I + h_k L_{s_k}) Z_k.$$

By (49), (50), and (54), the last two terms together have norm at most

$$C\big[h_k|Z_k|^2 + h_k^2|Z_k| + h_k^3\big].$$

This vector can be split into a part of norm at most $U_k = Ch_k(|Z_k| + h_k)|Z_k|$ and one of norm at most $V_k = Ch_k^3$. Explicitly, multiply it by $U_k/(U_k + V_k)$ and $V_k/(U_k + V_k)$, with the zero convention if both bounds vanish. For $Z_k \neq 0$, represent the first part by a rank-one operator $h_k E_k Z_k$ from $\mathcal{E}_{s_k}$ to $\mathcal{E}_{s_{k+1}}$; set $E_k = 0$ at $Z_k = 0$. We obtain the exact recurrence

$$\begin{aligned} Z_{k+1} &= A_k Z_k + h_k E_k Z_k + r_k, \\ |E_k| &\le C(|Z_k| + h_k) \to 0, \qquad |r_k| \le C_r h_k^3. \end{aligned} \tag{59}$$

The base and the sequences A_k, E_k, r_k are now frozen along the existing actual trajectory. This is a linear recurrence with fixed coefficient sequences for that trajectory, not a uniqueness statement for the nonlinear dynamics. Freezing assumes neither the weighted estimate to be proved nor center convergence.

3.4 The original-clock Green bootstrap

3.4.1 Bounded solutions and the next-fiber forcing index

Let X_∞ and X_2 be spaces of sections over the fixed discrete base sequence:

$$\|W\|_\infty = \sup_{k \ge J} |W_k|, \qquad \|W\|_2 = \sup_{k \ge J} \frac{|W_k|}{h_k^2}, \qquad W_k \in \mathcal{E}_{s_k}. \tag{60}$$

They are Banach spaces and X_2 embeds continuously in X_∞. No global trivialization is needed.

For forcing $f_j \in \mathcal{E}_{s_{j+1}}$, define

$$\begin{aligned} (\mathcal{G}f)_k = &\sum_{j=J}^{k-1} \Phi(k, j+1) \mathsf{P}^-_{s_{j+1}} f_j \\ &- \sum_{j=k}^{\infty} \Phi(k, j+1) \mathsf{P}^+_{s_{j+1}} f_j. \end{aligned} \tag{61}$$

The projections act at the forcing's *next* base s_{j+1}. In a moving bundle this index is essential. The sums converge absolutely when $|f_j| \le Ch_j$, by (58) and the divergent slow clock $\sum h_j = \infty$.

Every bounded actual solution of $W_{k+1} = A_k W_k + f_k$ satisfies

$$W_k = \Phi(k, J) \mathsf{P}^-_{s_J} W_J + (\mathcal{G}f)_k. \tag{62}$$

The stable part is forward variation of constants. For the unstable part, solve backward from N and use

$$|\Phi(k, N) \mathsf{P}^+_{s_N} W_N| \le C_A e^{-a(t_N - t_k)} \sup_{j \ge J} |W_j| \longrightarrow 0.$$

Thus unstable modes are treated by a justified backward terminal limit. They are not set to zero or silently dropped.

3.4.2 Uniform kernels with the original integer weights

There is a fixed finite C_G such that, after a preliminary late index with $h_J \le a/2$,

$$\begin{aligned} \|\mathcal{G}(hEW)\|_\infty &\le C_G \sup_{j\ge J} |E_j|\, \|W\|_\infty, \\ \|\mathcal{G}(hEW)\|_2 &\le C_G \sup_{j\ge J} |E_j|\, \|W\|_2, \\ \|\mathcal{G}r\|_2 &\le C_G C_r \quad \text{if } |r_j| \le C_r h_j^3. \end{aligned} \tag{63}$$

Here $(hEW)_j = h_j E_j W_j$ lies in the next fiber. The kernels from (58), including the $j+1$ offset, are bounded by a constant times $e^{-a|t_k - t_j|}$: the offset changes the exponent by at most ah_j.

For any fixed $\beta > 0$, the sums

$$\sum_{j\ge J} h_j e^{-\beta|t_k - t_j|}$$

are uniformly bounded in k. Compare each summand with the integral on the adjacent clock interval $[t_j, t_{j+1}]$, whose length is h_j, and use $h_j \le h_J$ to bound the exponential variation on that interval.

For the weighted bounds, when $j < k$,

$$\frac{h_j^2}{h_k^2} = \frac{k}{j}, \qquad \log\frac{k}{j} \le \sum_{m=j}^{k-1} \frac{1}{m} \le h_J \sum_{m=j}^{k-1} h_m = h_J(t_k - t_j). \tag{64}$$

The last inequality follows from $1/m = h_m^2 \le h_J h_m$. This consumes at most half of the exponential decay when $h_J \le a/2$. For $j \ge k$, the weight ratio is at most one. The same clock-interval comparison proves all of (63). In particular a forcing of order h_j^3 is h_j times the weight h_j^2.

The homogeneous stable section

$$H_k = \Phi(k, J)\mathsf{P}^-_{s_J} Z_J \tag{65}$$

also lies in X_2: (64) absorbs the factor h_j^2/h_k^2 into the exponential stable decay.

The order of constants is important. First choose a preliminary late index for the moving contraction bounds and $h \le a/2$. Freeze a, C_A and a bound C_G valid for all later suffixes, using that preliminary maximal step. Only then enlarge J to make $\sup_{k\ge J} |E_k|$ small. Enlarging J cannot worsen this fixed kernel bound. Changing the origin of t_k changes no clock differences and never changes $h_k = k^{-1/2}$.

3.4.3 Two-space identification of the actual section

Apply (62) to the actual bounded Z in (59):

$$Z = H + \mathcal{G}r + \mathcal{G}(hEZ). \tag{66}$$

Enlarge J so that $C_G \sup_{k\geq J} |E_k| < 1/2$. The affine map on the right is a contraction on X_∞ by the first bound in (63). It is also a contraction on X_2 by the second bound, and $H + \mathcal{G}r \in X_2$. The X_2 fixed point belongs to X_∞, and hence equals the actual bounded section by uniqueness in X_∞. Therefore

$$|Z_k| = O(h_k^2). \tag{67}$$

Using (46) gives

$$\operatorname{dist}_M(p_k, S) + |v_k| + |g(p_k)| = O(k^{-1}). \tag{68}$$

This proves the transverse and entire velocity rates before one-point convergence. The actual weighted norm was not assumed finite, and no infinite weighted norm was absorbed into itself.

3.5 Finite actual travel before one-point convergence

Insert $|v_k| = O(h_k^2)$ into the exact current-base identity (4) and the normal tracking estimate (8):

$$x_k - x_{k-1} = h_k v_k - \tfrac{1}{2} h_k^2 \mathrm{II}_{p_k}(v_k, v_k) + O(h_k^3).$$

The first and last terms are $O(h_k^3)$, and the curvature term is $O(h_k^6)$. Thus

$$|x_k - x_{k-1}| = O(k^{-3/2}). \tag{69}$$

Summing over the actual integer index proves finite total travel of (x_k) and its $O(k^{-1/2})$ travel tail. Any finite initial portion contributes only finite length. Hence (x_k) is Cauchy and converges to one point $p_* \in S$. Since $n_k \to 0$, the projected sequence (p_k) has the same limit. This is the first point in the proof at which convergence to one point is asserted.

The center estimate also now follows without any circular premise. Equations (50) and (67) imply

$$\operatorname{dist}_S(s_{k+1}, s_k) = O(h_k^3). \tag{70}$$

The sequence (s_k) has finite intrinsic travel in compact S, converges to p_*, and has intrinsic travel tail $O(k^{-1/2})$. Since $\operatorname{dist}_M(p_k, s_k) = O(h_k^2)$, this gives

$$\operatorname{dist}_M(p_k, p_*) + |x_k - p_*| = O(k^{-1/2}).$$

The stronger transverse statement remains valid: $n_k = O(h_k^2)$ and (68) imply $\operatorname{dist}(x_k, S) = O(k^{-1})$.

On the compact controlled neighborhood, ∇F is Lipschitz. The literal predictor in (2) gives

$$\begin{aligned} |y_k - x_k| &\leq |1 - \gamma h_k|\, |x_k - x_{k-1}| + b|\nabla F(x_k) - \nabla F(x_{k-1})| \\ &\leq C|x_k - x_{k-1}| = O(k^{-3/2}). \end{aligned}$$

Therefore $y_k \to p_*$, $\operatorname{dist}(y_k, S) = O(k^{-1})$, and $|y_k - p_*| = O(k^{-1/2})$. Moreover,

$$|y_{k+1} - y_k| \le |x_{k+1} - x_k| + |y_{k+1} - x_{k+1}| + |y_k - x_k| = O(k^{-3/2}).$$

This proves finite predictor travel and its $O(k^{-1/2})$ tail. It completes Theorems 2 and 3.

A center increment of order h_k^3 has a tail of order h_k, even though the transverse position and full scaled velocity are $O(h_k^2)$. The argument therefore does not upgrade the pointwise tail to $O(k^{-1})$ when $\dim S > 0$. If S is a point, there is no center position, and (67) directly gives $\operatorname{dist}_M(p_k, p_*) = O(k^{-1})$; the normal and predictor bounds give the same rate for x_k and y_k.

4 Actual bias and sharp center tails

For ambient vectors a, c, the contraction $D^3F(p)[a, c] \in \mathbb{R}^d$ is defined by

$$\langle z, D^3F(p)[a, c]\rangle = D^3F(p)[z, a, c], \qquad z \in \mathbb{R}^d.$$

All inverse normal Hessians act only on the normal space. Define on M

$$\nu(p) = N_p p, \qquad a(p) = -H_p^{-1}\nu(p), \qquad \psi(p) = \tfrac{1}{2}\langle \nu(p), H_p^{-1}\nu(p)\rangle, \tag{71}$$

$$\mathcal{T}(p) = P_p D^3F(p)[a(p), a(p)]. \tag{72}$$

Theorem 4 (Full vector bias at a Morse limit)**.** *Assume* (1) *and* (5)*. Suppose $q|_M$ is Morse, or, more locally, the admitted trajectory has a nondegenerate critical cluster point p_*. Write*

$$Q_* = \operatorname{Hess}_M q(p_*), \qquad a_* = -H_{p_*}^{-1}p_*, \qquad \beta_* = -\tfrac{1}{2}Q_*^{-1}\mathcal{T}(p_*), \qquad B_* = a_* + \beta_*.$$

Then p_* is the unique limit and the literal iterates satisfy

$$x_k = p_* + \frac{B_*}{k} + O(k^{-3/2}), \qquad y_k = p_* + \frac{B_*}{k} + O(k^{-3/2}). \tag{73}$$

Separately,

$$p_k = p_* + \frac{\beta_*}{k} + O(k^{-3/2}), \quad n_k = \frac{a_*}{k} + O(k^{-3/2}), \quad v_k = O(k^{-3/2}). \tag{74}$$

The Hessian Q_* need not be positive definite. An approached saddle or maximum is included. In zero tangent dimension, $\beta_* = 0$ and tangent assertions are vacuous.

Let $\tau_k : T_{p_k}M \to T_{p_{k+1}}M$ be parallel transport along the short geodesic. The normalized raw tangent-phase defect has the ambient limit

$$h_k^{-3}\{\tau_k^{-1}v_{k+1} - v_k + h_k[\gamma v_k + bg(p_k)]\} \longrightarrow -\tfrac{1}{8}\mathcal{T}(p_*). \tag{75}$$

In the graph coordinate $\sigma = P_{p_*}(p - p_*)$, $\omega = P_{p_*}v$, the normalized position defect tends to zero:

$$h_k^{-3}(\sigma_{k+1} - \sigma_k - h_k\omega_k) \longrightarrow 0. \tag{76}$$

For the Morse–Bott component use the notation K_s, V_s, Q_s and the moving coordinates of Section 3.

$$\Xi(s) = (Q_s|_{V_s})^{-1} P_{V_s} \operatorname{grad}_M \psi(s), \qquad W(s) = \tfrac{1}{6} \operatorname{grad}_S \psi(s). \tag{77}$$

An empty transverse space has $\Xi = 0$.

Theorem 5 (Center drift and moving-base bias)**.** *Assume* (1) *and* (5)*, and suppose every connected component of* $\operatorname{Crit}_M q$ *is compact and Morse–Bott. Alternatively, for the localized conclusion, it is enough that the omega-limit set of this history meets one compact connected Morse–Bott component* S *of* $\operatorname{Crit}_M q$*. There are a selected component* S *and* $p_* \in S$ *to which* x_k, y_k *converge.*

On a late suffix let s_k *be the intrinsic tubular projection of* p_k *to* S*, write* $p_k = \exp^M_{s_k} \eta_k$*,* $\eta_k \in V_{s_k}$*, and transport* v_k *back to* s_k *along that short normal geodesic to obtain* w_k*. Then*

$$\eta_k = h_k^2 \Xi(s_k) + O(h_k^3), \qquad w_k = h_k^2 W(s_k) + O(h_k^3), \qquad k v_k \longrightarrow W(p_*). \tag{78}$$

The fast normal variables obey

$$n_k = h_k^2 a(p_k) + O(h_k^4), \qquad u_k = O(h_k^4). \tag{79}$$

The point tails have the common leading term

$$\begin{aligned} s_k &= p_* - 2h_k W(p_*) + O(h_k^2), \\ x_k &= p_* - 2h_k W(p_*) + O(h_k^2), \qquad y_k = p_* - 2h_k W(p_*) + O(h_k^2). \end{aligned} \tag{80}$$

In particular

$$\sqrt{k}(x_k - p_*),\ \sqrt{k}(y_k - p_*) \longrightarrow -\tfrac{1}{3} \operatorname{grad}_S \psi(p_*). \tag{81}$$

Relative to the moving critical base,

$$\begin{aligned} x_k - s_k &= h_k^2 [a(s_k) + \Xi(s_k)] + O(h_k^3), \\ y_k - s_k &= h_k^2 [a(s_k) + \Xi(s_k)] + O(h_k^3), \end{aligned} \tag{82}$$

so both normalized differences tend to $-H_{p_*}^{-1} p_* + \Xi(p_*)$ *after multiplication by* k*.*

The actual increments and total-travel tails satisfy

$$k^{3/2}(x_k - x_{k-1}) \longrightarrow W(p_*), \qquad k^{3/2}(y_{k+1} - y_k) \longrightarrow W(p_*), \tag{83}$$

$$\begin{aligned} \sqrt{k} \sum_{j=k+1}^{\infty} |x_j - x_{j-1}| &\longrightarrow 2|W(p_*)|, \\ \sqrt{k} \sum_{j=k}^{\infty} |y_{j+1} - y_j| &\longrightarrow 2|W(p_*)|. \end{aligned} \tag{84}$$

These conclusions hold also when the leading constant vanishes.

All asymptotic constants above may depend on the admitted history. The uniformity over prepared entries needed for sharpness will be proved separately in Section 4.8. The moving-base limit (82) is not a full second-order coefficient about p_*. If $W(p_*) \neq 0$, $k(x_k - p_*)$ has no finite vector limit. Even after subtracting the leading term in (80), only an $O(k^{-1})$ fixed-point remainder is asserted. If S is a point, the center is fixed, $W = 0$, and (82) reduces to the full Morse bias. If $\dim S > 0$ and $W(p_*) = 0$, the improved $O(k^{-1})$ point bound alone does not identify its whole coefficient.

Theorem 6 (Actual sharpness)**.** *Let $M = \mathbb{S}^1$ and, on a fixed annulus about it, let*

$$F(r, \varphi) = \tfrac{1}{2}H(\varphi)(r-1)^2, \qquad H(\varphi) = 2 + \tfrac{1}{2}\sin\varphi. \tag{85}$$

Put $p_0 = (1, 0)$. Every sufficiently late original integer K admits the independent prepared history

$$x_{K-1} = x_K = \left(1 - \frac{1}{KH(0)}\right) p_0. \tag{86}$$

It satisfies (5) *with $B_0 = 0$. Constants independent of K give, for all $k \geq K$, $|n_k| + |u_k| + |v_k| \leq C/k$ and total lifted angular travel from k at most $C/\sqrt{k}$. For sufficiently large K the limiting angle obeys $|\varphi_*| < \pi/4$, and*

$$\sqrt{k}(x_k - p_*) \longrightarrow \frac{H'(\varphi_*)}{6H(\varphi_*)^2}(-\sin\varphi_*, \cos\varphi_*) \neq 0. \tag{87}$$

Consequently the $k^{-1/2}$ point and actual-travel exponents in this class are sharp.

Theorems 1, 2 and Corollary 3.1 supply all prior dynamical inputs. They give $n = h^2 a(p) + O(h^3)$, $u = O(h^3)$, and the estimates needed below, with the normal metric of Lemma 2 on the entire band $L < 8$. The full fiber is $\mathcal{E}_s = V_s \oplus T_s M$, and its corrected transport and slow dichotomy are those already constructed in Section 3.3. All asymptotic suffixes retain the original clock and may depend on the admitted history.

$$|p_k - p_*| + |v_k| = O(h_k^2), \qquad |x_k - x_{k-1}| + |y_k - x_k| = O(h_k^3) \tag{88}$$

in the Morse case, while in the Morse–Bott case

$$\begin{aligned} \operatorname{dist}_M(p_k, S) + |v_k| + |g(p_k)| &= O(h_k^2), \\ \operatorname{dist}_S(s_k, p_*) = O(h_k), \qquad & \operatorname{dist}_S(s_{k+1}, s_k) = O(h_k^3). \end{aligned} \tag{89}$$

4.1 The leading force from the literal two-line map

The two settings have the common late bounds

$$n = h^2 a(p) + O(h^3), \quad u = O(h^3), \quad |v| + |g(p)| = O(h^2), \quad d = x - x_{\text{prev}} = hv + u = O(h^3). \tag{90}$$

Here and below a missing index means the current k. The ambient Hessian H_p annihilates T_pM and equals H_p on N_pM. Taylor expansion, or integration of the Hessian along the segment of length d, gives

$$\nabla F(p+n) - \nabla F(p+n-d) = \mathsf{H}_p d + O((|n|+|d|)|d|) = H_p u + O(h^5). \tag{91}$$

Thus its tangent part is $O(h^5)$. Substitution into the exact predictor yields

$$y - p = n + hv + A_p u - \gamma h^2 v - \gamma h u + O(h^5). \tag{92}$$

In particular

$$P_p(y-p) = hv - \gamma h^2 v + O(h^5), \qquad y - p = h^2 a(p) + O(h^3). \tag{93}$$

Expanding the gradient at the actual y around the current p gives

$$\nabla F(y) = \mathsf{H}_p(y-p) + \tfrac{1}{2} D^3 F(p)[y-p, y-p] + O(|y-p|^3).$$

The Hessian term is normal. The cross term between $h^2 a(p)$ and the $O(h^3)$ remainder is $O(h^5)$, and the cubic gradient remainder is $O(h^6)$. Therefore

$$P_p \nabla F(y) = \tfrac{1}{2} h^4 \mathcal{T}(p) + O(h^5), \qquad h^2 P_p y = h^2 g(p) + O(h^5). \tag{94}$$

Write the next literal increment as $d_+ = x_{k+1} - x_k = \alpha + c$, with $\alpha = P_p d_+$ and $c = N_p d_+$. The second algorithm line now gives

$$\alpha = hv - \gamma h^2 v - b h^2 g(p) - \tfrac{b}{2} h^4 \mathcal{T}(p) + O(h^5). \tag{95}$$

Neither the gradient-history term nor regularization at y contributes an additional phase-independent term of order h^4. The normal increment is $c = O(h^3)$ by the already established actual history bound at $k+1$; directly its apparent h^2 term cancels because $H_p a(p) + \nu(p) = 0$.

4.1.1 Projection and transport

Put $p_+ = \pi(x_{k+1})$ and $\delta = \log_p p_+$. The exact representation is $x_{k+1} = p + \alpha + (n+c)$. Since $\pi(p+\zeta) = p$ for normal ζ, Taylor expansion in α gives

$$\delta = \alpha + O(|n+c||\alpha| + |\alpha|^2) = \alpha + O(h^5). \tag{96}$$

Consequently $|\delta| = O(h^3)$ and $\delta = hv + O(h^4)$. Let $\tau : T_pM \to T_{p_+}M$ be short-geodesic parallel transport. Then

$$\tau^{-1} P_{p_+}(\alpha + c) = \alpha + O(|\delta||c| + |\delta|^2|\alpha|) = \alpha + O(h^6). \tag{97}$$

For a tangent input, projection onto the next tangent space and parallel transport have the same first-order ambient normal term $\mathrm{II}_p(\delta, \alpha)$, leaving only a second-order discrepancy. A normal input contributes $O(|\delta||c|)$ by projector variation. Bounded local bundle derivatives make these estimates uniform. This calculation retains the current-base history convention.

4.1.2 The original clock

The exact ratio is

$$\frac{h_k}{h_{k+1}} = \sqrt{1+h_k^2} = 1 + \tfrac{1}{2}h_k^2 + O(h_k^4). \tag{98}$$

Divide (97) by h_{k+1} and use (95). The extra rescaling term $O(h^2 v)$ is $O(h^4)$, and the divided transport error is $O(h^5)$. Hence

$$\tau^{-1} v_{k+1} = v - h[\gamma v + bg(p)] - \tfrac{b}{2}h^3 \mathcal{T}(p) + O(h^4), \tag{99}$$

$$\log_p p_{k+1} = hv + O(h^4). \tag{100}$$

Thus the normalized tangent defect equals $-(b/2)\mathcal{T}(p) + O(h)$. The phase-independent position forcing at order h^3 is zero. These identities follow from literal increments; no stationary regularized branch has been substituted for the trajectory.

4.2 Compliance energy at critical bases

The forcing has a geometric interpretation at a critical base only.

Lemma 6. *At every $s \in \mathrm{Crit}_M q$,*

$$\mathrm{grad}_M \psi(s) = -\tfrac{1}{2}\mathcal{T}(s). \tag{101}$$

Proof. Differentiate $p = g(p) + \nu(p)$ in a tangent direction t. Taking normal components gives

$$\nabla^\perp_t \nu = -\mathrm{II}_p(t, g(p)).$$

At a critical s, this is zero for every $t \in T_s M$, including directions transverse to S inside M. Choose normal fields parallel for the normal connection at s. Their ambient derivatives at s are tangent, so the Hessian annihilates the extra differentiated-field terms. Therefore

$$\langle a, (\nabla^\perp_t H)a\rangle = D^3F(s)[t,a,a].$$

Differentiating (71) at s gives

$$\begin{aligned} d\psi_s[t] &= \langle \nabla^\perp_t \nu, H^{-1}\nu\rangle - \tfrac{1}{2}\langle H^{-1}\nu, (\nabla^\perp_t H)H^{-1}\nu\rangle \\ &= -\tfrac{1}{2}D^3F(s)[t,a,a]. \end{aligned}$$

This is (101). ∎

Away from the critical set, the first differentiated term need not vanish, so (101) is not used as an identity on all of M. At a critical base the velocity forcing becomes $b\,\mathrm{grad}_M \psi(s)$. Although q is constant on S, its component $b\,\mathrm{grad}_S \psi(s)$ can be nonzero.

For the Hessian sign convention, at a critical p_*,

$$\langle Q_*\xi, \zeta\rangle = \langle \xi, \zeta\rangle + \langle p_*, \mathrm{II}_{p_*}(\xi,\zeta)\rangle.$$

The convention makes the second fundamental form negative outward-normal-valued on a unit sphere. This also fixes the coefficient signs at saddles and maxima.

4.3 The shared Green operator at weight three

The finite-travel estimate gives $\operatorname{dist}_S(s_{k+1}, s_k) = O(h_k^3)$. Hence the corrected transport from (53) now satisfies $D_k - T_k = O(h_k^3)$, while the product maps $\Phi(k, l)$ retain (58). For a fixed nondegenerate point use $D_k = T_k = I$ and its fixed phase operator. The following weighted consequence of the same Green representation will be used in both cases.

Lemma 7 (Actual forced extraction at weight three)**.** *Let a bounded actual section satisfy*

$$E_{k+1} = A_k E_k + f_k, \qquad |f_k| \le C_f h_k^4,$$

with (58)*. Then* $E_k = O(h_k^3)$ *on a sufficiently late original suffix.*

Proof. The forcing f_j belongs to the next fiber $\mathcal{E}_{s_{j+1}}$. Variation of constants for a bounded solution gives

$$\begin{aligned} E_k = \Phi(k, J)\mathsf{P}^-_{s_J} E_J + \sum_{j=J}^{k-1} \Phi(k, j+1)\mathsf{P}^-_{s_{j+1}} f_j \\ - \sum_{j=k}^{\infty} \Phi(k, j+1)\mathsf{P}^+_{s_{j+1}} f_j. \end{aligned} \tag{102}$$

To justify the last sum, solve the unstable equation backward from N. Its terminal contribution is at most

$$C_A e^{-a(t_N - t_k)} \sup_{j \ge J} |E_j| \longrightarrow 0,$$

since $\sum h_j = \infty$. The minus sign and next-fiber index in (102) are essential. Thus an approached unstable component is handled by boundedness of its actual orbit, rather than setting an unstable coordinate to zero.

For $j < k$, the integer-clock weight obeys

$$\frac{h_j^3}{h_k^3} = \left(\frac{k}{j}\right)^{3/2}, \qquad \log \frac{k}{j} \le \sum_{m=j}^{k-1} \frac{1}{m} \le h_J(t_k - t_j).$$

Choose J so late that $(3/2)h_J \le a/2$. This ratio then consumes at most half the exponential decay in (58). For $j \ge k$, $h_j^3/h_k^3 \le 1$. The offset $j + 1$ changes the exponential estimate by a bounded factor. Finally,

$$\sup_{k \ge J} \sum_{j \ge J} h_j e^{-(a/2)|t_k - t_j|} < \infty,$$

by comparison with the integral over adjacent slow-clock intervals, of length h_j. Consequently the two forced sums are $O(h_k^3)$. The stable homogeneous term is also $O(h_k^3)$, since its exponential decay dominates every integer-clock power. This proves the lemma directly. There is no absorption of an a priori infinite weighted norm. ∎

4.4 Proof of the actual Morse bias

At the limit use the fixed graph coordinate

$$\sigma = P_{p_*}(p - p_*), \qquad \omega = P_{p_*}v, \qquad Z = (\sigma, \omega).$$

Locally $p = p_* + \sigma + \chi(\sigma)$ with χ normal at p_* and $\chi(0) = D\chi(0) = 0$. Thus $\chi(\sigma) = O(|\sigma|^2)$. Theorem 2 gives $|Z| = O(h^2)$.

The raw position identity (100), followed by the geodesic Taylor formula, gives

$$\sigma_{k+1} - \sigma_k = h_k\omega_k + O(h_k^4).$$

Transport changes a tangent vector of size $O(h^2)$ in ambient coordinates by $O(|\delta|h^2) = O(h^5)$. Since

$$P_{p_*}g(p) = Q_*\sigma + O(|\sigma|^2), \qquad P_{p_*}\mathcal{T}(p) = \mathcal{T}(p_*) + O(|\sigma|),$$

(99) yields the actual recurrence

$$Z_{k+1} = (I + h_kJ)Z_k + h_k^3C_* + r_k, \qquad r_k = O(h_k^4), \tag{103}$$

where

$$J = \begin{pmatrix} 0 & I \\ -bQ_* & -\gamma I \end{pmatrix}, \qquad C_* = (0, -\tfrac{b}{2}\mathcal{T}(p_*)).$$

Every eigenvalue of Q_* is real and nonzero, and the corresponding phase roots solve $r^2 + \gamma r + b\lambda = 0$. Hence J is hyperbolic even at an approached saddle or maximum and when a stable root is repeated.

The algebraic equation $JL_* + C_* = 0$ has the unique solution

$$L_* = (\beta_*, 0), \qquad \beta_* = -\tfrac{1}{2}Q_*^{-1}\mathcal{T}(p_*).$$

This alone does not prove tracking. For the actual $E_k = Z_k - h_k^2L_*$, (103) gives

$$E_{k+1} = (I + h_kJ)E_k + \widetilde{r}_k, \qquad \widetilde{r}_k = r_k + (h_k^2 - h_{k+1}^2)L_* = O(h_k^4),$$

using $h_k^2 - h_{k+1}^2 = 1/[k(k+1)]$. The already known $Z_k = O(h_k^2)$ ensures boundedness of this actual E. Lemma 7 proves

$$Z_k = h_k^2(\beta_*, 0) + O(h_k^3). \tag{104}$$

Returning through the graph chart gives $p_k = p_* + h_k^2\beta_* + O(h_k^3)$. The map $v \mapsto P_{p_*}v$ on $T_{p_k}M$ has uniformly bounded inverse near the limit, so $v_k = O(h_k^3)$. Equation (8) and smoothness of a give $n_k = h_k^2a_* + O(h_k^3)$. Adding these proves (73) for x.

Local Lipschitz continuity of ∇F and the exact predictor give $|y_k - x_k| \le C|x_k - x_{k-1}| = O(h_k^3)$, so y has the same coefficient. Equations (99) and (103) also prove (75) and (76).

In zero tangent dimension the projected base is eventually p_* and normal tracking alone gives the result. This proves Theorem 4.

By Lemma 6, the tangent coefficient may also be written $\beta_* = Q_*^{-1} \operatorname{grad}_M \psi(p_*)$. The full vector is generally not its normal part a_*. A first-order position expansion alone cannot be differentiated to obtain a k^2 increment coefficient; no such coefficient or optimization-residual asymptotic is asserted here.

4.5 Proof of the moving phase coefficient

Use the coordinates of Theorem 5, with $Z_k = (\eta_k, w_k) \in \mathcal{E}_{s_k}$. The finite-travel input gives $|Z_k| = O(h_k^2)$ and base motion $O(h_k^3)$. The full velocity, including K_s directions, is retained in w.

Taylor expansion of g at s_k, with transport to s_k, gives

$$g(p_k) \text{ transported to } s_k = Q_{s_k}\eta_k + O(|\eta_k|^2).$$

Changing the evaluation of $\mathcal{T}(p_k)$ to s_k costs $O(|\eta_k|)$, hence only $O(h_k^5)$ in the velocity step. The nonlinear phase-coordinate change costs $O(h_k|Z_k|^2) = O(h_k^5)$, and differentiation of a moving frame multiplies a base step $O(h_k^3)$ by a fiber vector $O(h_k^2)$. The remaining raw finite-step errors are $O(h_k^4)$ by (99)–(100). Thus the sharper actual recurrence is

$$T_k^{-1} Z_{k+1} = (I + h_k L_{s_k}) Z_k + h_k^3 C(s_k) + O(h_k^4), \qquad C(s) = (0, b \operatorname{grad}_M \psi(s)). \tag{105}$$

All statements are in a finite collection of compact bundle charts; no choice of global eigenvectors is needed.

The unique smooth algebraic coefficient is

$$\zeta(s) = -L_s^{-1} C(s) = (\Xi(s), W(s)). \tag{106}$$

Indeed $P_V W = 0$, and the second row separates into

$$bQ_s \Xi = bP_{V_s} \operatorname{grad}_M \psi, \qquad \gamma W = bP_{K_s} \operatorname{grad}_M \psi.$$

Since $b/\gamma = 1/6$, these are exactly (77). To prove actual extraction, set $e_k = Z_k - h_k^2 \zeta(s_k)$. Smoothness and the base-motion bound give

$$T_k^{-1}\big[h_{k+1}^2 \zeta(s_{k+1})\big] - h_k^2 \zeta(s_k) = O(h_k^4), \tag{107}$$

where the clock change is $-h_k^4/(1 + h_k^2)$ and the transported section change contributes $O(h_k^5)$. Subtracting the target from (105) yields

$$e_{k+1} = T_k(I + h_k L_{s_k}) e_k + f_k, \qquad f_k = O(h_k^4).$$

We already know $e_k = O(h_k^2)$. Replacing T_k by D_k changes the right side by $O(h_k^3)|e_k| = O(h_k^5)$. Therefore the actual section satisfies the exact equation

$$e_{k+1} = A_k e_k + r_k, \qquad r_k = O(h_k^4). \tag{108}$$

Here r_k is the actual difference; it may depend on the trajectory. Lemma 7 applies to this bounded section and proves $e_k = O(h_k^3)$. This proves (78), with stable and backward unstable contributions both included.

Finally $s_k - p_* = O(h_k)$, so smoothness gives $W(s_k) = W(p_*) + O(h_k)$. The ambient change of a transported tangent vector of magnitude $O(h_k^2)$ over the transverse distance $O(h_k^2)$ is $O(h_k^4)$. Hence

$$v_k = h_k^2 W(p_*) + O(h_k^3). \tag{109}$$

No conclusion about normalized actual increments is drawn yet: the previously known $u_k = O(h_k^3)$ is of exactly the order of $h_k v_k$. It must be improved first.

4.6 Refined fast-normal tracking

With $d = O(h^3)$, $n = O(h^2)$, and $u = O(h^3)$, the raw calculation gives

$$N_p(y - p) = n + (A_p - \gamma h)u + O(h^5),$$

$$N_p \nabla F(y) = H_p[n + (A_p - \gamma h)u] + O(h^4), \qquad h^2 N_p y = h^2 \nu(p) + O(h^4).$$

The geodesic chord has normal coordinate $O(|\delta|^2) = O(h^6)$, and base or normal-transport changes cost $O(|\delta|(|n| + |u|)) = O(h^5)$. Writing bars for normal backtransport to the current base, both actual rows are therefore

$$\begin{aligned} \bar{n}_{k+1} &= A_p n + A_p^2 u - b h^2 \nu(p) + O(h^4), \\ \bar{u}_{k+1} &= (A_p - I)n + A_p^2 u - b h^2 \nu(p) + O(h^4). \end{aligned} \tag{110}$$

The term $-\gamma h A_p u$ is $O(h^4)$ and has deliberately remained in the error. The relation $\bar{u}_{k+1} = \bar{n}_{k+1} - n + O(h^5)$ verifies the second row using the new current-base history, without replacing u by an old-base normal difference.

The vector $(a(p), 0)$ solves exactly

$$(a, 0) = B_p(a, 0) - b(\nu, \nu). \tag{111}$$

In addition, smoothness, $\delta = O(h^3)$ and the clock identity give

$$\text{backtransport}\big[h_{k+1}^2 (a(p_{k+1}), 0)\big] - h_k^2 (a(p_k), 0) = O(h_k^4).$$

Subtract this target from the actual recurrence (110). The normal contraction norm in Lemma 2, together with the $O(h_k^3)$ base variation, gives on a late suffix

$$|z_{k+1} - h_{k+1}^2 (a(p_{k+1}), 0)|_{\mathcal{Q}} \le \rho |z_k - h_k^2 (a(p_k), 0)|_{\mathcal{Q}} + C h_k^4, \quad z_k = (n_k, u_k),$$

for one fixed $\rho < 1$. A geometric convolution of $h_j^4 = j^{-2}$ is $O(h_k^4)$. Explicitly, for $j = k - 1 - \ell \ge J$, $(k/j)^2 \le (\ell + 2)^2$, and $\sum_{\ell \ge 0} (\ell + 2)^2 \rho^\ell < \infty$; the homogeneous entry term is also $O(h_k^4)$. This proves (79). The target is an algebraic approximation tracked by the actual sequence, not an assumed invariant normal graph.

4.7 Point tails and exact travel constants

The derivative of the base projection at $(\eta, w) = (0, 0)$ is $P_{K_s} w$. Equations (100) and (78) thus give

$$\log^S_{s_k} s_{k+1} = h_k P_{K_{s_k}} w_k + O(h_k^4) = h_k^3 W(s_k) + O(h_k^4).$$

Passing to ambient chords costs $O(h_k^6)$. Since $W(s_k) = W(p_*) + O(h_k)$,

$$s_{k+1} - s_k = h_k^3 W(p_*) + O(h_k^4). \tag{112}$$

The original integer sums satisfy

$$\sum_{j=k}^{\infty} j^{-3/2} = 2k^{-1/2} + O(k^{-3/2}), \qquad \sum_{j=k}^{\infty} j^{-2} = O(k^{-1}). \tag{113}$$

Sum (112) toward the already known limit p_* to get

$$s_k - p_* = -2h_k W(p_*) + O(h_k^2).$$

The ambient exponential expansion and (78) give

$$p_k = s_k + h_k^2 \Xi(s_k) + O(h_k^3).$$

Adding the refined normal estimate, with $a(p_k) = a(s_k) + O(h_k^2)$, proves the x moving-base formula (82) and the point tail (80).

To obtain literal increments, use the exact current-base identity (4), not a differentiation of the point expansion. Equations (109) and (79) yield

$$x_k - x_{k-1} = h_k^3 W(p_*) + O(h_k^4). \tag{114}$$

The gradient-difference estimate (91) now has $H_p u = O(h^4)$, so the actual predictor satisfies

$$y_k - x_k = h_k^3 W(p_*) + O(h_k^4). \tag{115}$$

This proves the same point and moving-base conclusions for y. Subtracting its offsets at consecutive indices is legitimate with the displayed errors: the difference of the explicit $h_k^3 W(p_*)$ terms is $O(h_k^5)$, and the two errors together are $O(h_k^4)$. Therefore

$$y_{k+1} - y_k = h_k^3 W(p_*) + O(h_k^4). \tag{116}$$

Taking norms in (114)–(116), summing and using (113) proves both constants in (84). This argument also covers $W(p_*) = 0$. It proves Theorem 5.

The transverse coefficient in (82) cannot be substituted for a fixed-point second-order coefficient. Determining a limit of $k[x_k - p_* + 2h_k W(p_*)]$ would require the next center term and the curvature of S. None is claimed here.

4.8 Uniform actual sharpness on a variable-stiffness circle

We give the entry-uniform existence argument, rather than presuming that a conditional nonzero coefficient is realized. For this section take any fixed smooth positive periodic H with $0 < \mu \le H \le L < 8$, and put

$$p(\varphi) = (\cos\varphi, \sin\varphi), \qquad t(\varphi) = (-\sin\varphi, \cos\varphi), \qquad F(r,\varphi) = \tfrac{1}{2}H(\varphi)(r-1)^2.$$

The model satisfies (1) on a fixed annulus. Here $S = M$, $q|_M = 1/2$, V_s is empty, and

$$a(p) = -\frac{p}{H(\varphi)}, \qquad \Xi = 0, \qquad \psi(\varphi) = \frac{1}{2H(\varphi)}, \qquad W(\varphi) = -\frac{H'(\varphi)}{12H(\varphi)^2}t(\varphi). \tag{117}$$

4.8.1 A uniform normal bound from the original entry

For any initial angle φ_0 prescribe

$$x_{K-1} = x_K = \left(1 - \frac{h_K^2}{H(\varphi_0)}\right) p(\varphi_0). \tag{118}$$

Then $p_K = p(\varphi_0)$, $n_K = -h_K^2 p(\varphi_0)/H(\varphi_0)$, and $u_K = v_K = 0$. Uniformly in φ_0, this satisfies (5) with $B_0 = 0$ once K is sufficiently large. Fix the capture caps and constants for this one B_0 before enlarging K.

Write $Z_k = |(n_k, u_k)|_{\mathcal{Q}_{p_k}}$ for the normal-pair norm of Lemma 2. The collision-safe one-step inequality (22) gives

$$Z_k \le C_0 h_K^2 \rho^{k-K} + C_N \sum_{i=K}^{k-1} \rho^{k-1-i} h_i^2.$$

Both terms are uniformly $O(h_k^2)$. Indeed, for $j = k - K$ and $\ell = k - 1 - i$,

$$\frac{h_K^2 \rho^j}{h_k^2} = \frac{k}{K}\rho^j \le (1+j)\rho^j, \qquad \frac{h_i^2}{h_k^2} = \frac{k}{i} \le \ell + 2.$$

Uniform norm equivalence and the geometric sums therefore provide one $M_N < \infty$, independent of K, φ_0, such that

$$|n_k| + |u_k| \le M_N h_k^2 \quad \text{for every } k \ge K. \tag{119}$$

There is no $h_K \rho^{k-K}$ transient to wait out: the prepared normal state is already $O(h_K^2)$. The capture theorem also provides a common velocity cap $|v_k| \le R$.

4.8.2 A pointwise tangent inequality on the fixed caps

At a current step write

$$n_k = \zeta p, \qquad u_k = \upsilon p, \qquad v_k = \omega t, \qquad |\zeta| + |\upsilon| \le M_N h^2, \quad |\omega| \le R.$$

The predecessor is exactly

$$x_{k-1} = (1 + \zeta - \upsilon)p - h\omega t.$$

The explicit ambient gradient is

$$\nabla F(r, \varphi) = H(\varphi)(r-1)p(\varphi) + \frac{H'(\varphi)}{2r}(r-1)^2 t(\varphi). \tag{120}$$

For $z = (1+\eta)p + at$ with $|\eta| \le Ch^2$, $|a| \le Ch$, elementary polar geometry gives $|r_z - 1| \le Ch^2$ and $|\varphi_z - \varphi_k| \le C|a|$. Projecting (120) onto $t = t(\varphi_k)$ yields

$$|\langle t, \nabla F(z)\rangle| \le C(|a|h^2 + h^4). \tag{121}$$

The radial gradient contributes $O(|a|h^2)$, and the intrinsic tangent gradient contributes $O(h^4)$. All constants depend only on the already fixed caps and model.

Applying this to x_k, x_{k-1} and then the literal predictor gives

$$y_k = (1+\eta_y)p + a_y t, \qquad \eta_y = O(h^2), \qquad a_y = h\omega - \gamma h^2\omega + O(h^3|\omega| + h^4).$$

Applying (121) to y_k is legitimate. Its Tikhonov tangent term is exactly $h^2 a_y t$, since $P_p p = 0$. The corrector thus has

$$x_{k+1} = (1+\eta_+)p + at, \qquad \eta_+ = O(h^2), \qquad a = h\omega - \gamma h^2\omega + O(h^3|\omega| + h^4). \tag{122}$$

For a sufficiently small fixed maximal step $1 + \eta_+ > 1/2$. Choose the unique small angular increment

$$\delta_k = \arctan\frac{a}{1+\eta_+}, \qquad |\delta_k| \le C(h|\omega| + h^4).$$

This defines consistent lifted angles recursively; they need not be assumed to remain in any chosen angular chart.

Put $c = \eta_+ - \zeta = O(h^2)$. Since $x_{k+1} - x_k = at + cp$ and $t(\varphi_{k+1}) = (\cos\delta_k)t - (\sin\delta_k)p$, the next scalar velocity is exactly

$$\omega_{k+1} = \frac{a\cos\delta_k - c\sin\delta_k}{h_{k+1}}.$$

Its numerator differs from a by at most

$$C(|a|\delta_k^2 + |c||\delta_k|) \le C(h^3|\omega| + h^6).$$

Using the exact clock (98) therefore proves

$$\omega_{k+1} = (1 - \gamma h_k)\omega_k + \varepsilon_k, \qquad |\varepsilon_k| \le C_V(h_k^2|\omega_k| + h_k^3). \tag{123}$$

The phase-independent forcing is $O(h_k^3)$, not $O(h_k^2)$. No eventual one-point convergence or asymptotic velocity rate was used in deriving this entry-uniform inequality.

4.8.3 A shrinking barrier and finite angular travel

Set $a_0 = \gamma/2$. Enlarge one fixed entry threshold so $C_V h_K \le \gamma/2$, $\gamma h_K \le 1$, and $h_K \le a_0/2$. Then

$$|\omega_{k+1}| \le (1 - a_0 h_k)|\omega_k| + C_V h_k^3.$$

Choose once and for all $A_V \ge \max\{1, 2C_V/a_0\}$. The initial velocity is zero. If $|\omega_k| \le A_V h_k^2$, then

$$\begin{aligned} |\omega_{k+1}| &\le A_V h_k^2 - (a_0 A_V - C_V) h_k^3 \\ &\le A_V h_k^2 - \tfrac{1}{2} a_0 A_V h_k^3 \le A_V h_k^2 - A_V h_k^4 \le A_V \frac{h_k^2}{1 + h_k^2} = A_V h_{k+1}^2. \end{aligned}$$

The last inequality uses $1 - t \le (1 + t)^{-1}$ for $t \ge 0$. Finite induction proves $|v_k| \le A_V h_k^2$ for every $k \ge K$, with constants independent of the final entry index.

The angular increment bound now gives $|\delta_k| \le C_\delta(h_k^3 + h_k^4) \le 2C_\delta k^{-3/2}$. Since

$$\sum_{j=k}^{\infty} j^{-3/2} \le k^{-3/2} + 2k^{-1/2} \le 3k^{-1/2},$$

we obtain

$$\sum_{j=k}^{\infty} |\varphi_{j+1} - \varphi_j| \le C_* k^{-1/2}, \qquad |\varphi_* - \varphi_0| \le C_* K^{-1/2}. \tag{124}$$

This also proves existence of the angular limit and, with (119), of the actual limit $p(\varphi_*)$. The constants were fixed in the order: model and capture caps; normal contraction and M_N; pointwise tangent constant C_V; barrier and angular constants; then the common entry threshold. None depends on the selected limit or a history-dependent late onset.

4.8.4 A nonzero coefficient for a fixed smooth model

Now specialize to (85) and $\varphi_0 = 0$. Its normal spectrum is $[3/2, 5/2] \subset (0, 4)$. Choose any original K above the preceding common threshold with $C_* K^{-1/2} < \pi/4$. Equation (124) forces $|\varphi_*| < \pi/4$, so $H'(\varphi_*) = \frac{1}{2}\cos\varphi_* > 0$.

Apply the already proved Theorem 5 to this actual admitted history and use (117). This gives (87); its coefficient is genuinely nonzero. The point distance is asymptotic to a positive constant times $k^{-1/2}$, and the travel constants in (84) are positive as well. Thus no universal $o(k^{-1/2})$ improvement holds in this class. This proves Theorem 6. The construction uses sufficiently late independent histories, not an assertion that an arbitrary run begun at index one reaches them.

4.9 A nonzero tangent Morse coefficient

The translated-circle objective

$$F(X, Y) = \frac{1 + 0.2Y}{8}\big((X - 2)^2 + Y^2 - 1\big)^2$$

has at $p_* = (1, 0)$ the values $H_* = 1$, $Q_* = 2$, $a_* = (-1, 0)$, $\mathcal{T}(p_*) = (0, 0.2)$, and therefore $\beta_* = (0, -0.05)$ and $B_* = (-1, -0.05)$. The normal spectrum on the circle is $[0.8, 1.2]$. The tangent coefficient is thus genuinely present. The accompanying exact checks verify these tensors; finite trajectories in the data supplement illustrate, rather than prove, the asymptotics.

At the same critical base $p = (1, 0)$, consider the independent one-step input $x = x_{\mathrm{prev}} = (1 - h^2, 0)$. Although $v = u = g(p) = 0$, direct substitution into the literal map gives the exact raw tangent increment in the direction $(0, -1)$:

$$\langle (0, -1), x_+ - x \rangle = \frac{0.2}{8} h^4 (1 + h^2/2)^2. \tag{125}$$

For this family of one-step inputs as $k \to \infty$, the next scaled tangent velocity, divided by h^3, tends to 0.025. Thus the additive phase-independent defect is genuinely present even at a critical phase. This exact identity is distinct from an infinite-time assertion about an orbit.

5 Finite actual travel through degenerate critical sets

Definition 1 (Concave one-sided gradient inequality)**.** A C^1 function f on a Riemannian manifold has the concave Kurdyka–Łojasiewicz property at a point a if there are a neighborhood U of a, a number $\eta > 0$, and a function $\varphi : [0, \eta) \to [0, \infty)$, continuous and concave with $\varphi(0) = 0$, of class C^1 on $(0, \eta)$ and with $\varphi' > 0$, such that

$$\varphi'\big(f(x) - f(a)\big) \, \| \operatorname{grad} f(x) \| \geq 1 \quad \text{for } x \in U, \quad 0 < f(x) - f(a) < \eta.$$

The property is one-sided; no local-minimum assumption is included.

Let Ω be the nonempty compact omega-limit set of (p_k) supplied by capture. Its criticality follows from the same result. The proof below obtains its common q level independently of any gradient inequality or one-point convergence.

Theorem 7 (Finite actual travel from the scalar inequality)**.** *Assume* (1) *and* (5)*. Suppose that q has the concave Kurdyka–Łojasiewicz property at every point of Ω. Then there is one point $p_* \in \mathrm{Crit}_M \, q$ such that*

$$p_k \longrightarrow p_*, \qquad x_k \longrightarrow p_*, \qquad y_k \longrightarrow p_*. \tag{126}$$

Moreover,

$$\sum_{k=K}^{\infty} h_k \big(|v_k| + |g(p_k)| + h_k^2 \big) < \infty, \tag{127}$$

$$\sum_{k=K+1}^{\infty} |x_k - x_{k-1}| < \infty, \qquad \sum_{k=K}^{\infty} |y_{k+1} - y_k| < \infty. \tag{128}$$

For some $k_0 \geq K$,

$$\sum_{k=k_0}^{\infty} \operatorname{dist}_M(p_{k+1}, p_k) < \infty. \tag{129}$$

No isolatedness, Hessian nondegeneracy, or manifold structure of $\operatorname{Crit}_M q$ *or* Ω *is assumed. The limit can be a saddle or a maximum of* q*.*

5.1 The full energy criterion and quantitative tails

Write $Q_p = \operatorname{Hess}_M q(p)$ and $L_q = \sup_M \|Q_p\|$. Fix

$$0 < \varepsilon \leq \frac{\gamma}{2\big(L_q + \gamma^2/(2b)\big)}. \tag{130}$$

On $\mathcal{N} = TM \times \mathbb{R}$, with the Sasaki metric on TM and the Euclidean metric in the last coordinate, define

$$\begin{aligned} \mathcal{E}(p, v) &= bq(p) + \tfrac{1}{2}|v|^2 + \varepsilon\langle g(p), v\rangle, \\ \mathcal{H}(p, v, z) &= \mathcal{E}(p, v) + Az^3, \qquad X_k = (p_k, v_k, h_k). \end{aligned} \tag{131}$$

The Sasaki metric makes the Levi-Civita horizontal and vertical subspaces orthogonal and uses the base metric on each. This is the same mechanical energy used for capture; its admissible ε bound agrees with Section 2. The augmentation constant $A > 0$ is a scalar, unrelated to the normal operator A_p. Both constants are fixed, never step-dependent.

Theorem 8 (Full augmented energy criterion)**.** *Under* (1) *and* (5)*, for each fixed choice* (130) *there is a finite $A_0 > 0$, allowed to depend on the admitted history, such that for every $A \geq A_0$, $\mathcal{H}(X_k)$ is strictly decreasing on a sufficiently late suffix and has a finite limit ℓ. Before any point convergence is known,*

$$\widehat{\Omega} = \Omega \times \{0\} \times \{0\}, \qquad \mathcal{H}|_{\widehat{\Omega}} = \ell, \qquad q|_{\Omega} = q_{\Omega} := \ell/b, \tag{132}$$

and $\widehat{\Omega}$ is the compact critical omega-limit set of (X_k).

If $\mathcal{H}$ has the concave Kurdyka–Łojasiewicz property at every point of $\widehat{\Omega}$, all conclusions of Theorem 7 hold. Let Φ be any uniform concave desingularizer for $\mathcal{H}$ near this set and level, and write $e_k = \mathcal{H}(X_k) - \ell$. For all sufficiently late k,

$$e_k > 0, \qquad \sum_{j=k}^{\infty} h_j\big(|v_j| + |g(p_j)| + h_j^2\big) \leq C\Phi(e_k). \tag{133}$$

Define the actual travel tail by

$$\begin{aligned} \mathcal{L}_k = &\sum_{j=k+1}^{\infty} |x_j - x_{j-1}| + \sum_{j=k}^{\infty} |y_{j+1} - y_j| \\ &+ \sum_{j=k}^{\infty} \operatorname{dist}_M(p_{j+1}, p_j), \end{aligned} \tag{134}$$

Then

$$\mathcal{L}_k \le C\Phi(e_k), \qquad |x_k - p_*| + |y_k - p_*| + \operatorname{dist}_M(p_k, p_*) \le C\Phi(e_k). \tag{135}$$

The scalar hypothesis of Theorem 7 implies this full-energy hypothesis. If φ_q is a uniform scalar desingularizer at the level q_Ω, an explicit admissible choice is, on a small positive interval,

$$\Phi(t) = C_\Phi \left[3b\varphi_q\!\left(\frac{t}{3b}\right) + 2\sqrt{t} + 3t^{1/3} \right], \tag{136}$$

with a sufficiently large fixed C_Φ.

Corollary 5.1 (Analytic and definable geometry)**.** *Every trajectory admitted by* (5) *has the conclusions of Theorems 7 and 8 if either*

1. *M is real analytic embedded, or*
2. *M is C^5 and definable in a fixed o-minimal expansion of the real field.*

In either case F may remain an arbitrary C^5 function satisfying (1)*. Analyticity or definability of F is not required. In the analytic case a power desingularizer is available; arbitrary o-minimal definability guarantees a general desingularizer and is not asserted to give a power exponent.*

Corollary 5.2 (Power bounds at the original clock)**.** *In Theorem 7, suppose there are $c_q > 0$, $\eta_q > 0$, and $0 \le \vartheta_q < 1$ such that, near Ω,*

$$|g(p)| \ge c_q\big(q(p) - q_\Omega\big)^{\vartheta_q} \quad \text{whenever } 0 < q(p) - q_\Omega < \eta_q. \tag{137}$$

Set $\vartheta = \max\{\vartheta_q, 2/3\}$. Then

$$e_k = O\big(k^{-1/[2(2\vartheta-1)]}\big), \qquad \mathcal{L}_k = O\big(k^{-(1-\vartheta)/[2(2\vartheta-1)]}\big). \tag{138}$$

Each distance in (135) *has the same upper bound as $\mathcal{L}_k$. In particular $\vartheta_q \le 2/3$ gives $e_k = O(k^{-3/2})$ and $\mathcal{L}_k = O(k^{-1/2})$. These are upper bounds, not claims of optimal actual-trajectory rates or pointwise velocity and increment asymptotics.*

5.2 Inputs from the common capture engine

Set $V_k = |v_k|$, $G_k = |g(p_k)|$, $T_k = V_k^2 + G_k^2$, and $s_k = V_k + G_k$; missing indices denote the current step. Theorem 1 gives bounded phase, vanishing v, g, the retained normal transient and the current-base normal-history expansion. Lemma 3 gives the phase-sensitive raw defect without any Morse–Bott hypothesis. Compact M has finitely many connected components, since connected coordinate neighborhoods admit a finite subcover. Their positive separation and the vanishing projected steps put p_k eventually in one component, with a uniform short-geodesic radius. Throughout this section $Q_p = \operatorname{Hess}_M q(p)$ denotes the intrinsic tangent Hessian; it is distinct from the fast normal-pair metric of Lemma 2.

$$h_{k+1} = h_k(1 + h_k^2)^{-1/2}, \qquad h_k = k^{-1/2}. \tag{139}$$

5.3 A strict mechanical energy with a cubic clock

We first prove the descent and common-level part of Theorem 8, without any Kurdyka–Łojasiewicz assumption.

5.3.1 Horizontal and vertical derivatives

For the function $\mathcal{E}$ in (131), its horizontal and vertical gradients are exactly

$$D_h\mathcal{E} = bg + \varepsilon Qv, \qquad D_v\mathcal{E} = v + \varepsilon g. \tag{140}$$

A horizontal derivative parallel transports v, so the kinetic term has zero horizontal derivative and the cross term contributes εQv. Consequently, on the compact phase region,

$$\|\operatorname{grad}\mathcal{E}\| \le C_E(V + G). \tag{141}$$

For the leading phase field $f(p, v) = (v, -\gamma v - bg)$, the derivative computation in (25) gives

$$d\mathcal{E}[f] \le -\tfrac{\gamma}{2}V^2 - \tfrac{\varepsilon b}{2}G^2 \le -a_0 T, \qquad a_0 = \min\{\gamma/2, \varepsilon b/2\} > 0. \tag{142}$$

The cross term is signed. We do not require $\mathcal{E} - bq(p_*)$ to be nonnegative near a critical point. The captured compact disk bundle is enough to bound $\mathcal{E}$ above and below along the trajectory.

5.3.2 The discrete Taylor remainder

Write $\delta = \log_p p_+$ and $w = \tau^{-1}v_+ - v$. Join the two actual phase points by

$$p(t) = \exp_p(t\delta), \qquad v(t) = \tau_{p\to p(t)}(v + tw), \qquad 0 \le t \le 1.$$

Its initial derivative gives

$$d\mathcal{E}[\delta, w] = b\langle g, \delta\rangle + \langle v, w\rangle + \varepsilon\big(\langle Q\delta, v\rangle + \langle g, w\rangle\big). \tag{143}$$

The derivatives of q through order three are bounded. A second derivative along this path consists of bounded multiples of $|\delta|^2$, $|w|^2$, $|\delta||w|$, and the term involving $(\nabla Q)[\delta, \delta, v(t)]$. Thus compactness gives a uniform constant C_T with

$$\mathcal{E}(p_+, v_+) - \mathcal{E}(p, v) = d\mathcal{E}[\delta, w] + R_E, \qquad |R_E| \le C_T(|\delta| + |w|)^2. \tag{144}$$

This uses finite-chart bounds over all bases, not coordinates around an assumed limit.

By (35) and $s \le \sqrt{2T}$,

$$|(\delta, w) - hf(p, v)| \le C(h^2\sqrt{T} + h^3), \qquad |\delta| + |w| \le C(h\sqrt{T} + h^3).$$

Combining (141), (142), and (144), there are fixed finite $C_1, C_2, C_3 \ge 0$ such that

$$\mathcal{E}_{k+1} - \mathcal{E}_k \le -a_0 h_k T_k + C_1 h_k^2 T_k + C_2 h_k^3\sqrt{T_k} + C_3 h_k^6. \tag{145}$$

Choose a later suffix with $h_k \le 1$ and $C_1 h_k \le a_0/4$. Young's inequality gives

$$C_2 h_k^3 \sqrt{T_k} \le \tfrac{a_0}{4} h_k T_k + \tfrac{C_2^2}{a_0} h_k^5.$$

Hence, with $B = C_2^2/a_0 + C_3$,

$$\mathcal{E}_{k+1} - \mathcal{E}_k \le -\tfrac{a_0}{2} h_k T_k + B h_k^5. \tag{146}$$

A bare phase-independent $O(h_k^2)$ velocity error would instead cost $O(h_k^3)$ after absorption. The refined literal defect is what makes the cubic-clock compensation possible.

5.3.3 Fixed augmentation and the gradient upper bound

The mean-value theorem applied to (139) gives, for $0 < h_k \le 1$,

$$h_{k+1}^3 - h_k^3 = -\tfrac{3}{2} h_k^5 (1 + \xi_k h_k^2)^{-5/2} \le -d_0 h_k^5, \qquad d_0 = \frac{3}{2 \cdot 2^{5/2}}, \tag{147}$$

where $0 < \xi_k < 1$. Fix

$$A \ge A_0 := \frac{B+1}{d_0}. \tag{148}$$

For $S_k = V_k + G_k + h_k^2$, equations (146)–(148) imply

$$\begin{aligned} \mathcal{H}(X_k) - \mathcal{H}(X_{k+1}) &\ge \tfrac{a_0}{2} h_k T_k + h_k^5 \\ &\ge c_0 h_k (T_k + h_k^4) \ge c_d h_k S_k^2, \end{aligned} \tag{149}$$

where $c_0 = \min\{a_0/2, 1\}$ and $c_d = c_0/3$. The last step uses $S_k^2 \le 3(T_k + h_k^4)$. This is strict at every late step, even when $v_k = g(p_k) = 0$.

The full gradient is

$$\| \operatorname{grad} \mathcal{H}(p, v, z) \|^2 = |bg + \varepsilon Q v|^2 + |v + \varepsilon g|^2 + 9A^2 z^4. \tag{150}$$

In particular,

$$\| \operatorname{grad} \mathcal{H}(X_k) \| \le C_H S_k, \qquad C_H = \max\{b + \varepsilon, 1 + \varepsilon L_q, 3A\}. \tag{151}$$

The direction of this estimate is important: it is an upper bound. It will combine with the lower gradient inequality to yield a lower bound on the derivative of the desingularizer.

The actual augmented phase also has a length scale $h_k S_k$. Along the exponential/transport path above the Sasaki speed is $(|\delta_k|^2 + |w_k|^2)^{1/2}$. Adding the clock segment and using $|h_{k+1} - h_k| \le h_k^3/2$ gives

$$\operatorname{dist}_{\mathcal{N}}(X_{k+1}, X_k) \le C h_k S_k. \tag{152}$$

The cubic is signed for $z < 0$ and unbounded below on the whole z axis. This is harmless: the actual sequence lies in a compact subset of $TM \times [0, h_{k_0}]$, so $\mathcal{H}(X_k)$ is bounded below there. By (149) it decreases to a finite ℓ.

5.4 The common level before point convergence

Compactness, (7), and $h_k \to 0$ show that the omega-limit set of X_k is exactly

$$\widehat{\Omega} = \{(p, 0, 0) : p \in \Omega\}.$$

If $p \in \Omega$, continuity of $\mathcal{H}$ and convergence of its values give

$$bq(p) = \mathcal{H}(p, 0, 0) = \ell. \tag{153}$$

Thus $q = q_\Omega = \ell/b$ on Ω. This common-level fact is proved by energy convergence, not inferred merely from connectedness of a smooth critical set. Also $g = 0$ on Ω, so (150) shows that every point of $\widehat{\Omega}$ is critical for $\mathcal{H}$.

Both $\operatorname{dist}_M(p_k, \Omega) \to 0$ and $\operatorname{dist}_{\mathcal{N}}(X_k, \widehat{\Omega}) \to 0$. Otherwise a subsequence staying a fixed distance away would, by compactness, have a cluster point outside the corresponding cluster set. The intrinsic and coordinate distances are locally equivalent on the compact regions in use. No assumption about the dimension, smoothness, or number of points of Ω is needed.

Write $e_k = \mathcal{H}(X_k) - \ell$. Strict decrease implies $e_k > 0$: a zero gap would force the next value below the limit. More explicitly, finite summation of (149), followed by the limit, gives

$$e_k \geq \sum_{j=k}^{\infty} h_j^5 > 0. \tag{154}$$

This also prevents a spurious finite-termination case caused by the auxiliary variable.

5.4.1 Uniformization on a compact constant level

For later use, consider any C^1 function f which is constant on a compact set C and satisfies Definition 1 at each point of C. Choose finitely many of its local neighborhoods covering C, and restrict all associated desingularizers φ_i to the smallest positive level interval. The sum

$$\varphi = \sum_i \varphi_i$$

is continuous and concave, vanishes at zero, and has positive derivative. At a point of any chosen neighborhood, φ' dominates the derivative needed for its local inequality. A smaller fixed metric neighborhood of C lies in the finite union. This proves a uniform inequality there.

This is the finite-cover mechanism of the uniformized inequality in Bolte, Sabach, and Teboulle [3, Lemma 3.6, pp. 20–21]; their displayed lemma is Euclidean. The argument just given applies intrinsically once the local manifold inequalities are available. Neither connectedness of C nor a lower bound for f on the entire manifold is required.

Applying this argument to $\mathcal{H}$ at $\widehat{\Omega}$ gives a uniform Φ whenever the hypothesis of Theorem 8 holds. Applying it to q at Ω gives φ_q under Theorem 7. The common-level proof (153) precedes both applications.

5.5 Transferring the scalar inequality to the full energy

We now prove the implication asserted in Theorem 8. The argument is local near the compact constant critical level and uses no analytic or definable structure.

Lemma 8 (One-sided scalar to augmented energy transfer)**.** *Let $q \in C^3(M)$, let Ω be compact with $g = 0$ and $q = q_\Omega$ on Ω, and suppose that near Ω, $\|Q\| \le L_q$ and*

$$\varphi_q'\big(q(p) - q_\Omega\big)|g(p)| \ge 1 \quad \text{if } 0 < q(p) - q_\Omega < \eta_q, \tag{155}$$

for a concave desingularizer φ_q. Fix $b, A > 0$ and $\varepsilon > 0$ with $\varepsilon^2 L_q \le b/2$. Then the function (131) *has a uniform concave Kurdyka–Łojasiewicz inequality at level $\ell = bq_\Omega$ near $\widehat{\Omega}$. Formula* (136) *is an admissible desingularizer for a sufficiently large C_Φ. The neighborhood is open in $TM \times \mathbb{R}$, allowing $z < 0$ and $q(p) < q_\Omega$.*

Proof. Write $s = |v| + |g|$ and denote the phase-gradient components by

$$D_h = bg + \varepsilon Q v, \qquad D_v = v + \varepsilon g.$$

Their block identity is

$$D_h - \varepsilon Q D_v = (bI - \varepsilon^2 Q)g. \tag{156}$$

Since Q is self-adjoint, the smallest eigenvalue of $bI - \varepsilon^2 Q$ is at least $b/2$. Therefore

$$|g| \le \frac{2}{b}\big(|D_h| + \varepsilon L_q |D_v|\big), \qquad |v| \le |D_v| + \varepsilon |g|.$$

There is a fixed $c_E > 0$ with $\|\operatorname{grad}\mathcal{E}\| \ge c_E s$. Combining this with the clock component $3Az^2$ in (150) gives

$$\|\operatorname{grad}\mathcal{H}\| \ge c(s + z^2), \qquad c = \frac{\min\{c_E, 3A\}}{\sqrt{2}} > 0. \tag{157}$$

This lower bound and (151) play different roles.

Put $t = \mathcal{H} - bq_\Omega > 0$ and $C_0 = 1 + \varepsilon$. Irrespective of either sign in the lemma,

$$t \le b(q - q_\Omega)_+ + C_0 s^2 + A|z|^3. \tag{158}$$

The positive part only increases the scalar term; $|z|^3$ only increases the cubic; and the signed cross term is at most $\varepsilon|g||v|$. At least one term on the right is at least $t/3$.

Shrink the base neighborhood so that $|q - q_\Omega| < \eta_q$, and restrict $0 < t < 3b\eta_q$. For (136),

$$\Phi'(t) = C_\Phi \left[\varphi_q'\left(\frac{t}{3b}\right) + t^{-1/2} + t^{-2/3}\right]. \tag{159}$$

We check the three possible dominant terms separately.

If $b(q - q_\Omega)_+ \ge t/3$, then $q - q_\Omega > 0$ and $t/(3b) \le q - q_\Omega$. Concavity makes φ_q' nonincreasing, so

$$\varphi_q'\left(\frac{t}{3b}\right) \ge \varphi_q'(q - q_\Omega).$$

Equations (155), (157), and (159) imply

$$\Phi'(t)\|\operatorname{grad}\mathcal{H}\| \geq C_\Phi c.$$

This is the only case in which the scalar inequality is invoked.

If $C_0 s^2 \geq t/3$, then $s \geq \sqrt{t/(3C_0)}$ and the middle term of (159) yields

$$\Phi'(t)\|\operatorname{grad}\mathcal{H}\| \geq \frac{C_\Phi c}{\sqrt{3C_0}}.$$

Finally, if $A|z|^3 \geq t/3$, then $z^2 \geq [t/(3A)]^{2/3}$ and the last term yields

$$\Phi'(t)\|\operatorname{grad}\mathcal{H}\| \geq \frac{C_\Phi c}{(3A)^{2/3}}.$$

Thus it suffices to choose

$$C_\Phi \geq c^{-1}\max\{1, \sqrt{3C_0}, (3A)^{2/3}\}. \tag{160}$$

The function in (136) has all the required continuity, concavity, and positive-derivative properties. The base restrictions and compactness permit a fixed metric neighborhood of $\widehat{\Omega}$. This proves the full one-sided inequality on that neighborhood without restricting the signs of $q - q_\Omega$ or z. ∎

The energy choice already satisfies the block condition with room to spare. Set $a = \gamma^2/(2b)$. Since $(L_q + a)^2 \geq 4aL_q$, equation (130) gives

$$\varepsilon^2 L_q \leq \frac{\gamma^2 L_q}{4(L_q + a)^2} \leq \frac{\gamma^2}{16a} = \frac{b}{8} < \frac{b}{2}. \tag{161}$$

For $L_q = 0$ the first conclusion is immediate. No new restriction on the admitted history or on ε is introduced by the transfer.

The common level from Section 5.4, its finite-cover uniformization, and Lemma 8 prove the full-energy hypothesis from the scalar one. A neighborhood containing no strict upper scalar levels simply gives a vacuous scalar inequality; the other two cases of the transfer still control every positive augmented-energy gap.

5.6 Concavity gives actual finite travel

For the direct full-energy hypothesis or for its transferred version, we now have fixed $\eta, \delta > 0$ and a uniform concave Φ such that

$$\Phi'\big(\mathcal{H}(X) - \ell\big)\|\operatorname{grad}\mathcal{H}(X)\| \geq 1 \tag{162}$$

when $\operatorname{dist}_{\mathcal{N}}(X, \widehat{\Omega}) < \delta$ and $0 < \mathcal{H}(X) - \ell < \eta$. Section 5.4 ensures that X_k eventually lies in this set.

5.6.1 First-power phase summability

Concavity at the current energy gap gives

$$\Phi(e_k) - \Phi(e_{k+1}) \geq \Phi'(e_k)(e_k - e_{k+1}). \tag{163}$$

Equations (162) and (151) imply, in the needed direction,

$$\Phi'(e_k) \geq \frac{1}{\| \operatorname{grad} \mathcal{H}(X_k)\|} \geq \frac{1}{C_H S_k}.$$

The denominator is nonzero because $S_k \geq h_k^2 > 0$; the gradient itself is nonzero by (162). Combining with (149) and (163) yields

$$\Phi(e_k) - \Phi(e_{k+1}) \geq \frac{c_d}{C_H} h_k S_k. \tag{164}$$

Sum first over the finite interval $k \leq j \leq N$ and only then let $N \to \infty$. As $e_N \to 0$ and $\Phi(0) = 0$,

$$\sum_{j=k}^{\infty} h_j S_j \leq \frac{C_H}{c_d} \Phi(e_k) < \infty. \tag{165}$$

No unknown infinite quantity has been absorbed, and no convergence along a critical set has been assumed. Equation (152) also proves finite augmented-phase travel. More directly, (35) give

$$\operatorname{dist}_M(p_{k+1}, p_k) = |\delta_k| \leq C h_k S_k. \tag{166}$$

Thus p_k is Cauchy on its eventual compact connected component and converges to $p_* \in M$. Since $g(p_k) \to 0$, the limit is critical.

5.6.2 The actual normal history and ambient iterates

The expansion in (8), bounded II, and the exact identity (4) imply

$$|d_k| \leq h_k V_k + C h_k^2 V_k^2 + C h_k^3 \leq C h_k S_k. \tag{167}$$

For the last inequality use $h_k \leq 1$ and a fixed velocity cap $V_k \leq R$, so $h_k^2 V_k^2 \leq R h_k V_k$. Summing (167) using (165) proves finite actual x travel. Its limit equals p_* because $n_k \to 0$.

One can also check the normal portion directly: the capture dissipation and (8) already imply

$$\sum_k |u_k| \leq C \sum_k h_k^2 V_k^2 + C \sum_k h_k^3 < \infty.$$

The newly supplied first-power tangent sum is the missing part. Neither argument estimates the variation of an unspecified $O(h_k^3)$ normal-position remainder.

5.6.3 Predictor travel and tail indices

On the controlled neighborhood, the gradients along sufficiently short neighboring segments have a uniform Lipschitz constant L_F. The exact predictor formula gives

$$\begin{aligned} |y_k - x_k| &\le |1 - \gamma h_k||d_k| + b|\nabla F(x_k) - \nabla F(x_{k-1})| \\ &\le C_y|d_k|. \end{aligned} \tag{168}$$

Hence $y_k - x_k \to 0$ and $y_k \to p_*$. Successive predictors satisfy

$$|y_{k+1} - y_k| \le (1 + C_y)|d_{k+1}| + C_y|d_k|. \tag{169}$$

This proves finite actual y travel without a separate variation estimate for its time-dependent coefficient.

The x tail from x_k uses $d_{k+1}, d_{k+2}, \ldots$; the y tail additionally uses d_k through (169). Both are controlled by the phase tail starting at k. Equations (165), (166), and (167) therefore give exactly

$$\mathcal{L}_k \le C\Phi(e_k).$$

Taking the limit in each path bounds the distances to p_* by the same right side. No additive $O(h_k)$ remainder is needed: $\sum_{j\ge k} h_j^3$ is already included in (165). All omitted prefixes contain finitely many actual increments. Restoring them proves the original-index statements of Theorems 7 and 8.

5.7 Analytic and definable sufficient cases

We justify Corollary 5.1, including the metric transfer needed for the inequalities. In a relatively compact manifold chart, the positive-definite metric and its inverse have bounded eigenvalues away from zero. Intrinsic gradient norms and coordinate Euclidean gradient norms are consequently equivalent, uniformly after shrinking the chart. A finite cover supplies the uniform constants needed near a compact set.

5.7.1 Real analytic geometry

If M is real analytic embedded, $q = |p|^2/2$ and the induced metric are real analytic in local coordinates. The local analytic gradient inequality gives a power bound for q at every point of Ω. One precise source is Absil, Mahony, and Andrews [2, Lemma 2.1, p. 533]. Local metric equivalence gives the intrinsic inequality; Woodward [5, Appendix A, Theorem A.1 and the following paragraph, p. 48] explicitly records this manifold transfer. Neither source's separate gradient-flow or descent convergence theorem is imported.

A finite cover, restriction to scalar gaps below 1, and the largest of the finitely many exponents give (137). Theorems 7–8 and Corollary 5.2 apply. There is no requirement that the whole sequence enter one chart before uniformization.

The augmented analytic hypothesis can also be verified directly. In analytic base coordinates ξ and tangent fiber coordinates w, write the metric coefficients as $m_{ij}(\xi)$. Then

$$\mathcal{H}(\xi, w, z) = bq(\xi) + \tfrac{1}{2} m_{ij}(\xi) w^i w^j + \varepsilon\, \partial_i q(\xi) w^i + Az^3. \tag{170}$$

All coefficients are analytic; the cross term is simply $\varepsilon\, dq_\xi(w)$. Thus $\mathcal{H}$ is analytic on $TM \times \mathbb{R}$, and the same local inequality, metric comparison, and finite-cover argument give the full-energy hypothesis. The objective F and its normal Hessian do not occur in (170); their C^5 regularity has already served its purpose in capture and the raw estimates.

5.7.2 Definable geometry

Suppose M is C^5 and definable in one fixed o-minimal expansion of the real field. Definable local graph charts have definable derivatives; their tangent bundle, induced metric, q, and g are definable. Formula (170), now in these C^5 charts, makes $\mathcal{H}$ definable and C^1 with room to spare.

Kurdyka [4, Theorem 1, p. 775] supplies a differentiable definable positive-level desingularizer on a bounded definable open chart. Apply it to $\mathcal{H} - \mathcal{H}(a) > 0$. Its positive increasing desingularizer has a finite limit at zero; subtract that limit, normalize the inequality, and, if necessary, add a positive linear term. Definable one-variable monotonicity makes the derivative eventually monotone. If nonincreasing, the function is concave. If nondecreasing, dominate its derivative on a smaller interval by a fixed positive constant and use a linear desingularizer instead. The metric comparison above transfers the inequality, and Section 5.4 uniformizes it. The same argument can be applied to q. If no positive levels occur, the local inequality is vacuous.

This proves the definable case. Arbitrary o-minimal definability supplies a general desingularizer; the power-form refinements for more restrictive structures are not needed. The assumption is on the geometry, and F need not be definable.

5.8 Power exponents and original clock rates

Assume (137) and choose

$$\varphi_q(t) = \frac{t^{1-\vartheta_q}}{c_q(1-\vartheta_q)}.$$

For the transferred function (136),

$$\Phi'(t) = C_\Phi \left[c_q^{-1} \left(\frac{t}{3b} \right)^{-\vartheta_q} + t^{-1/2} + t^{-2/3} \right].$$

With $\vartheta = \max\{\vartheta_q, 2/3\}$ and $0 < t < \min\{1, \eta\}$, there is a fixed $D > 0$ such that $\Phi'(t) \le Dt^{-\vartheta}$. The full gradient inequality therefore implies

$$\|\operatorname{grad} \mathcal{H}(X)\| \ge D^{-1} (\mathcal{H}(X) - \ell)^{\vartheta}. \tag{171}$$

The derivative upper bound, not just a bound on $\Phi(t)$, justifies this replacement. A pure power desingularizer for (171) is

$$\Phi_{\mathrm{pow}}(t) = \frac{D}{1-\vartheta}\, t^{1-\vartheta}. \tag{172}$$

The floor 2/3 is forced by the full augmented neighborhood: for $p \in \Omega$, $v = 0$, and $z > 0$,

$$\mathcal{H}(p,0,z) - \ell = Az^3, \qquad \|\operatorname{grad}\mathcal{H}(p,0,z)\| = 3Az^2.$$

A power lower gradient bound can hold as $z \downarrow 0$ only if $3\vartheta \geq 2$. Thus the exponent of q cannot simply be reused when it is smaller.

Combining (171) with (151) gives $S_k \geq D^{-1}C_H^{-1}e_k^{\vartheta}$. The descent estimate yields the current-gap recurrence

$$e_{k+1} \leq e_k - ch_k e_k^{2\vartheta} \tag{173}$$

for a fixed $c > 0$. Let $r_\vartheta = 2\vartheta - 1 > 0$. Convexity of $t \mapsto t^{-r_\vartheta}$ gives

$$e_{k+1}^{-r_\vartheta} - e_k^{-r_\vartheta} \geq r_\vartheta e_k^{-r_\vartheta - 1}(e_k - e_{k+1}) \geq r_\vartheta c h_k.$$

Summing at the original clock,

$$e_k \leq \left[e_{k_0}^{-r_\vartheta} + r_\vartheta c \sum_{j=k_0}^{k-1} j^{-1/2} \right]^{-1/r_\vartheta}. \tag{174}$$

Since the sum is at least $2(\sqrt{k} - \sqrt{k_0})$, this is the energy bound in (138). The clock lower bound (154) has order $k^{-3/2}$, consistent with the upper bound when $\vartheta = 2/3$.

Use (172) in the already proved tail theorem. Then

$$\mathcal{L}_k \leq Ce_k^{1-\vartheta} = O\big(k^{-(1-\vartheta)/[2(2\vartheta-1)]}\big).$$

The same estimate holds for the distances and the weighted phase tail. This proves Corollary 5.2. Local power inequalities at individual points of Ω also suffice: a finite cover and gaps below 1 permit the largest exponent and a common positive constant.

6 Actual instability at the upper endpoint

The loss of strict frozen stability at $H = 8$ does not alone prove instability of a time-dependent algorithm. The next theorem concerns the exact matrices of the actual recursion.

Theorem 9 (Quadratic endpoint instability and normalized pair limit)**.** *Let $F(x) = 4x^2$ on $\mathbb{R}$ and $M = \{0\}$. In the literal map, keep $b = 1/4$ and $\gamma = 3/2$, and use either*

$$h_k = k^{-1/2}, \quad \varepsilon_k = h_k^2 = k^{-1}, \qquad \textit{or} \qquad h_k = k^{-1/3}, \quad \varepsilon_k = h_k^2 = k^{-2/3}.$$

For every sufficiently late original K and every nonzero history $Z_K = (x_K, x_{K-1})$, the globally defined actual trajectory satisfies

$$\frac{|Z_k|^2}{\prod_{j=K}^{k-1}(1+\gamma h_j)(1+bh_j^2)} \longrightarrow C(Z_K, K) \in (0, \infty). \tag{175}$$

Consequently $|Z_k| \to \infty$ and the scalar sequence (x_k) is unbounded. On the reciprocal schedule,

$$\log|Z_k| = \frac{3}{2}\sqrt{k} - \frac{7}{16}\log k + O(1), \tag{176}$$

whereas on the critical schedule,

$$\log|Z_k| = \frac{9}{8}k^{2/3} - \frac{21}{16}k^{1/3} + \frac{9}{16}\log k + O(1). \tag{177}$$

Both statements also give

$$\frac{\log|Z_k|}{\sum_{j=K}^{k-1} h_j} \longrightarrow \frac{\gamma}{2} = \frac{3}{4}. \tag{178}$$

For every $\theta > 0$ there are arbitrarily small nonzero histories at arbitrarily large K satisfying the same scaled normal-entry form as in Theorem 1 whose trajectories are unbounded.

Proof. The Hessian is 8, the normal kernel is trivial, and the singleton M is a compact smooth manifold. The map is globally defined. Its exact predictor is

$$y_k = -\gamma h_k x_k + (1+\gamma h_k)x_{k-1},$$

and its exact corrector is $x_{k+1} = -(1+bh_k^2)y_k$. Put

$$p(h) = \gamma h(1+bh^2), \qquad q(h) = (1+\gamma h)(1+bh^2).$$

Then

$$Z_{k+1} = T(h_k)Z_k, \qquad T(h) = \begin{pmatrix} p(h) & -q(h) \\ 1 & 0 \end{pmatrix}, \qquad \det T(h) = q(h) > 1. \tag{179}$$

Here $q(h)$ is a scalar multiplier, distinct from the restricted squared norm used earlier. Define

$$G(h) = \begin{pmatrix} 1 & -p(h)/2 \\ -p(h)/2 & q(h) \end{pmatrix}.$$

Direct multiplication gives the exact conformal identity

$$T(h)^* G(h) T(h) = q(h) G(h). \tag{180}$$

For $0 \le h \le 1/2$, one has $|p(h)| \le 51/64$ and $1 \le q(h) \le 119/64$. Row bounds imply

$$\tfrac{1}{2} I \le G(h) \le 3I.$$

The matrix G has bounded derivative on this interval and $G(0) = I$. Let $\delta_k = h_k - h_{k+1} > 0$. For one finite C,

$$(1 - C\delta_k)G(h_k) \le G(h_{k+1}) \le (1 + C\delta_k)G(h_k).$$

Choose K late enough that $C\delta_k \le 1/2$. The positive energy $E_k = Z_k^* G(h_k) Z_k$ therefore satisfies

$$(1 - C\delta_k)q(h_k)E_k \le E_{k+1} \le (1 + C\delta_k)q(h_k)E_k. \tag{181}$$

The clock variation telescopes: $\sum_{k \ge K} \delta_k = h_K$. Thus the two products of $1 \pm C\delta_k$ converge to positive finite values. Uniform metric equivalence gives

$$c_K |Z_K|^2 \prod_{j=K}^{k-1} q(h_j) \le |Z_k|^2 \le C_K |Z_K|^2 \prod_{j=K}^{k-1} q(h_j), \tag{182}$$

where $0 < c_K \le C_K < \infty$ are independent of the nonzero initial direction.

There is an actual limit as well. Invertibility of every $T(h_k)$ ensures $E_k > 0$ for a nonzero history. Set

$$\epsilon_k = \frac{E_{k+1}}{q(h_k)E_k} - 1.$$

Equation (181) gives $|\epsilon_k| \le C\delta_k$, so $\sum |\epsilon_k| < \infty$ and $\prod(1 + \epsilon_k)$ converges to a positive finite number. Hence $E_k / \prod_{j=K}^{k-1} q(h_j)$ has a positive finite limit. As $G(h_k) \to I$, the same is true with E_k replaced by $|Z_k|^2$, proving (175).

For large j, $\log q(h_j) \ge c h_j$ and $\sum h_j = \infty$. Thus the product diverges and $|Z_k| \to \infty$. A bounded scalar sequence would have bounded consecutive pairs, so (x_k) is unbounded. This argument does not assert that every individual $|x_k|$ tends to infinity; oscillation is allowed.

The exact logarithmic expansion is

$$\log q(h) = \frac{3}{2}h - \frac{7}{8}h^2 + \frac{9}{8}h^3 + O(h^4).$$

For $h_j = j^{-1/2}$ its cubic and higher terms are summable, while

$$\sum_{j=K}^{k-1} j^{-1/2} = 2\sqrt{k} + O(1), \qquad \sum_{j=K}^{k-1} j^{-1} = \log k + O(1).$$

Taking half of the logarithm of (182) proves (176). For $h_j = j^{-1/3}$, retain the cubic term and use

$$\sum_{j=K}^{k-1} j^{-1/3} = \tfrac{3}{2}k^{2/3} + O(1), \qquad \sum_{j=K}^{k-1} j^{-2/3} = 3k^{1/3} + O(1).$$

The fourth-order remainder is summable, giving (177). These sum estimates follow by summing the finite differences of the corresponding powers, whose errors are summable.

Finally $\log q(h_j) = \gamma h_j + O(h_j^2)$ and $\sum h_j^2 = o(\sum h_j)$: bound the tail by $h_j^2 \le \epsilon h_j$ and treat the finite prefix separately. This proves (178).

For the entry assertion take $x_{K-1} = x_K = \delta$ with $0 < |\delta| \le \theta h_K$. Then $n_K = \delta$, $u_K = 0$, and $v_K = 0$ because $T_0 M = \{0\}$. The history meets the normal-entry test and can be as small as desired, yet (175) forces unboundedness. ∎

Thus $L < 8$ has a sharp upper endpoint for a universal bounded-capture theorem with the stated independent-history entry. This is not a classification of every nonlinear objective beyond the endpoint, nor does it exclude the zero endpoint history.

7 Constant restricted norm and centered spheres

Suppose q is constant on a connected component M_0 of M. Take $S = M_0$. Then $V_s = \{0\}$, the phase bundle contains only $w \in T_s M_0$, and

$$L_s = -\gamma I.$$

It is entirely stable, with no unstable directions and no transverse position variables. The proof neither treats tangent positions as hyperbolic nor removes the damped tangent velocities. Since $g = 0$, (35) directly gives

$$|v_{k+1}| \le (1 - \gamma h_k + C h_k^2)|v_k| + C h_k^3.$$

The scalar stable convolution yields $|v_k| = O(h_k^2)$, and (8) with (4) gives finite actual travel. This is the rank-zero transverse case of the moving-bundle proof.

For an explicit literal model on the unit circle or unit sphere, take

$$F(x) = \frac{\lambda}{8}(|x|^2 - 1)^2, \qquad 0 < \lambda < 8. \tag{183}$$

Its gradient is $(\lambda/2)(|x|^2 - 1)x$, its normal Hessian on the unit sphere is λ, and $q|_M \equiv 1/2$. Write $\nabla F(x) = f(x)x$. The exact predictor and second step satisfy

$$\begin{aligned} y_k &= \big(2 - \gamma h_k - b f(x_k)\big)x_k + \big(-1 + \gamma h_k + b f(x_{k-1})\big)x_{k-1}, \\ x_{k+1} &= \sigma_k y_k, \qquad \sigma_k = 1 - b\big(f(y_k) + h_k^2\big). \end{aligned} \tag{184}$$

Thus the exterior-product history obeys the exact identity

$$x_k \wedge x_{k+1} = \sigma_k\big[1 - \gamma h_k - b f(x_{k-1})\big](x_{k-1} \wedge x_k). \tag{185}$$

In the late $O(h_k^2)$ normal regime, both $f(x_{k-1})$ and $f(y_k)$ are $O(h_k^2)$, so the multiplier is $1 - \gamma h_k + O(h_k^2)$. This checks the sign and the surviving tangent-velocity damping independently of the bundle notation. The radial symmetry may yield stronger special-case rates, from which no general rate is inferred.

Another permitted constant-q model, defined on an annulus, is

$$F(r,\varphi) = \tfrac{1}{2}H(\varphi)(r-1)^2, \qquad 0 < \mu \le H(\varphi) \le L < 8. \tag{186}$$

The unit circle is still the critical manifold and $q|_M$ is constant, but a nonconstant H removes rotational symmetry. It is useful for finite-step diagnostics of center drift.

For two transverse Hessian eigenvalues $\lambda_+ > 0$ and $\lambda_- < 0$, and two component-tangent velocity modes, (48) gives the exact fiber characteristic polynomial

$$(r+\gamma)^2(r^2+\gamma r+b\lambda_+)(r^2+\gamma r+b\lambda_-).$$

This explicit block example retains both the unstable transverse mode and the damped tangent velocities. The source checks also retain a structural moving-splitting diagnostic: for planar rotations R and natural transport equal to the identity, projector correction gives

$$D_k = \cos(\delta_k)R(\alpha_k+\delta_k)R(\alpha_k)^T.$$

The sequence $\delta_k = h_k/\log(k+10)$ has $\delta_k/h_k \to 0$ but divergent accumulated rotation. It illustrates why the proved dichotomy needs slow motion, not prior finite base length. This sequence is not claimed to be an IGATH trajectory. Exact checks of the fiber characteristic polynomial and the radial wedge identity accompany the finite-horizon circle data.

8 Scope and reproducibility

Every trajectory assertion is conditional on the original-index independent-history entry test. General asymptotic constants and sufficiently late proof suffixes may depend on that history; the prepared-circle estimates are explicitly uniform in the entry index. No constants are claimed uniform as $\mu \downarrow 0$, $L \uparrow 8$, or a transverse critical Hessian gap closes. The sharp interval concerns the fast normal block, while Morse–Bott and KL hypotheses constrain the separate tangent dynamics.

The scalar KL route permits degenerate critical sets and gives finite travel, but does not imply the per-step Morse–Bott rates or vector coefficients. The Morse–Bott route permits saddles and maxima, retains every velocity mode, and yields a generally nonzero summable center drift. Neither route asserts arbitrary index-one capture, generic saddle avoidance, global minimum-norm selection, an exact invariant normal graph, a complete second-order expansion around a positive-dimensional limit, or an optimization-residual asymptotic.

The critical schedule $h_k = k^{-1/3}$ has a different selection mechanism: its h_k^3 secondary drift is harmonic, unlike the summable drift here. Its separate capture and secondary-selection proof is developed in the companion manuscript [6]. The common sharp normal metric is Lemma 2; the endpoint result for both schedules is Theorem 9. Neither

reciprocal-clock capture nor its velocity barrier is imported into the critical schedule by changing an exponent.

The source package includes the editable integrated proof, the exact common style, a reproducible build, a complete result map, and mathematical checks and data. Symbolic identities and finite computations supplement the proofs; they do not certify existential capture thresholds or infinite-time behavior. Reports 33–36 and the reciprocal/core portions of report 39 are integrated here, with the earlier report 29 capture prerequisite replaced by the full sharp-range proof. The original narrower sufficient theorems remain valid specializations. The critical-clock portions of report 39 are routed to the companion. No priority or journal-acceptance assertion is made.

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